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Systems and methods for printing a flexible web and printing compositions

Abstract

A printing system comprises a transport apparatus adapted to transport a flexible web (24) along a process direction and first and second individually controllable inkjet imager units (44, 60, 70, 82) offset from one another along the process direction. Each of the first imager unit and the second imager unit includes a first portion (226) operable to print on a first portion of the web and a second portion (228) operable to print on the second portion of the web wherein each of the first portion and second portion of the first and second imager units is stationary along the process direction and the lateral direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application Ser. No. 62/988,723, filed Mar. 12, 2020 and entitled “Systems and Methods for Printing A Flexible Web and Printing Compositions,” the entirety of which is incorporated herein by reference. This application claims priority to U.S. Provisional Patent Application Ser. No. 62/988,731, filed Mar. 12, 2020 and entitled “Systems and Methods for Printing A Flexible Web and Printing Compositions,” the entirety of which is incorporated herein by reference.

BACKGROUND

(1) The present subject matter relates to printing systems and methods, and more particularly to systems and methods for printing a flexible web.

(2) Flexographic printing is a well-known technique in the printing industry utilizing a flexible relief plate. It is used for printing on many substrates, including plastic, metallic films, and other water impermeable material.

(3) Ink jet printing is also a well-known technique wherein a digital image is recreated upon a substrate by depositing droplets of ink onto the substrate.

(4) Additionally, high speed printing systems have been developed for printing on a substrate, such

as a web of shrinkable polymeric film. Such a material typically exhibits both elasticity and plasticity characteristics that depend upon one or more applied influences, such as force, heat, chemicals, electromagnetic radiation, etc. These characteristics must be carefully taken into account during the system design process because it may be necessary: 1.) to control material shrinkage during imaging so that the resulting imaged film may be subsequently used in a shrink-wrap process, and 2.) to avoid system control problems by minimizing dynamic interactions between system components due to the elastic deformability of the substrate. Such considerations also impact the process of registering printed content so that the content is accurately reproduced.

(5) Specifically, a flexible web may be printed simplex (i.e., on one side) or duplex (that is, two sided). In either event, separately printed images, even if printed by a single printing unit (e.g., a multi-color imager unit), must be accurately registered with one another to minimize misregistration errors, such as color shifts, moire, undesired dot gain effects, or the like.

(6) Furthermore, the use of water-based inks and coatings for commercial print applications, including but not limited to flexography and ink jet printing, has been on the increase due in part to environmental and health concerns about volatile organic compounds (“VOC’s”) in solvent-based compositions that emit during the drying process.

(7) Health concerns are highlighted when printing for the food industry. For instance, Switzerland has put legislation in place regulating food-packaging printing inks. A list of components that may be used in printing inks have been compiled in an effort to remove some substances deemed carcinogenic, mutagenic, or toxic to reproduction. While only required in Switzerland, the Swiss Ordinance RS 817.023.21 with respect to printing inks and coatings alike is generally accepted as useful when creating ink(s) and/or ink receptive compound(s) for indirect food contact purposes, such as for food packaging, worldwide.

(8) As for general printing on a substrate or web that is porous or permeable, water within the ink is partially absorbed by the surface of the web during a drying process. However, there exists a problem when water-based inks are deposited on a web that is impermeable, such as a plastic web, metal web, and similar surfaces. Since inks dry primarily via evaporation during a drying and/or curing period, the lack of ability of the water-based ink to penetrate or absorb into the web itself leads to individual ink droplets spreading across the surface of the web. If a compilation of individual ink droplets spread and touch one another, the desired image quality may be adversely affected due to coalescing of the adjacent ink droplets. This is a problem that typically occurs with high-speed printing.

(9) Additionally, another problem during high speed printing known as “ink retransfer” or “pickoff” may occur, where the ink for the printed image has not sufficiently dried before contacting another part of the web system, such as an idler roller, and the ink is transferred unintentionally from the printed area to the roller.

(10) Furthermore, if during the printing process the temperature of the substrate exceeds the substrate's threshold for dimensional integrity, the substrate may shrink and/or deform leading to unusable product.

(11) The discussion above is merely provided for general background information and is not intended to be used as an aid in determining the scope of the claimed subject matter.

BRIEF DESCRIPTION

(12) According to one aspect, a printing system comprises a transport apparatus adapted to transport a flexible web along a process direction and first and second individually controllable ink jet imager units offset from one another along the process direction. Each of the first imager unit and the second imager unit includes a first portion operable to print on a first portion of the web and a second portion operable to print on the second portion of the web wherein each of the first portion and second portion of the first and second imager units is stationary along the process direction and the lateral direction. The first imager unit prints on the web at a first resolution and the second imager unit prints on the web at a second resolution different than the first resolution. A

first closed-loop control system controls web handling and a distance of at least one of the first portion and the second portion of at least one of the first imager unit and the second imager unit from the web. A second closed-loop control system controls registration between the first imager unit and the second imager unit.

(13) According to another aspect, a printing system for printing a heat-shrinkable web comprises a transport apparatus adapted to transport a flexible web along a process direction and first and second individually controllable ink jet imager units offset from one another along the process direction. Each of the first imager unit and the second imager unit includes a first portion operable to print on a first portion of the web and a second portion operable to print on the second portion of the web wherein each of the first portion and second portion of the first and second imager units is stationary along the process direction and the lateral direction. The first imager unit includes printheads that print on the web at a first resolution and the second imager unit includes printheads that print on the web at a second resolution different than the first resolution. A first closed-loop control system controls web handling and a distance of at least one of the printheads from the web and a second closed-loop control system is responsive to web temperature and controls print drying without substantial shrinking of the web. A third closed-loop control system controls registration between the first imager unit and the second imager unit.

(14) According to yet another aspect, a printing system for printing a polymeric heat shrinkable web comprises a first imager unit for applying a first printing composition to the web, a second imager unit for applying a second printing composition to the web, and a closed-loop control system that controls web handling and a distance of at least one printhead of at least one of the first imager unit and the second imager unit from the web. The first printing composition comprises at least one of a polymer, a defoamer agent, a surfactant, a surface treatment agent, an opacifier, and water. The second printing composition comprises at least one of a pigment, a polymer, a co-solvent, a surfactant, and water.

(15) According to a still further aspect, a printing system for printing a polymeric heat shrinkable web comprises a first imager unit for applying a first printing composition to the web, a second imager unit for applying a second printing composition to the web, and a closed-loop control system that controls web handling and a distance of at least one printhead of at least one of the first imager unit and the second imager unit from the web. The first printing composition comprises at least one of a viscosity modifier, a polymer, a surfactant, a defoamer agent, a surface treatment agent, an antimicrobial agent, and water. The second printing composition comprises at least one of a pigment, a polymer, a co-solvent, a surfactant, and water.

(16) According to yet another aspect, a method of printing comprises the steps of transporting a flexible web along a process direction and providing first and second individually controllable ink jet imager units offset from one another along the process direction. Each of the first imager unit and the second imager unit includes a first portion operable to print on a first portion of the web and a second portion operable to print on the second portion of the web and each of the first portion and second portion of the first and second imager units is stationary along the process direction and the lateral direction. Further, the first imager unit is operable to print on the web at a first resolution and the second imager unit is operable to print on the web at a second resolution different than the first resolution. The method further includes the steps of controlling web handling, controlling a distance of at least one of the first portion and the second portion of at least one of the first imager unit and the second imager unit from the web, and controlling registration between the first imager unit and the second imager unit.

(17) According to a still further aspect, a method of printing a heat-shrinkable web comprises the steps of transporting a flexible web along a process direction and operating first and second individually controllable ink jet imager units offset from one another along the process direction. Each of the first imager unit and the second imager unit includes a first portion operable to print at a first resolution on a first portion of the web and a second portion operable to print at a second

resolution different than the first resolution on the second portion of the web. Further, each of the first portion and second portion of the first and second imager units is stationary along the process direction and the lateral direction and each of the first portions and each of the second portions includes at least one printhead. The method further includes the steps of controlling web handling, controlling a distance of at least one of the printheads from the web, controlling print drying web responsive to web temperature without substantial shrinking of the web, and controlling registration between the first imager unit and the second imager unit.

(18) According to yet another aspect, a method of printing a web of polymeric heat-shrinkable material comprises the steps of operating a first imager unit for applying a first printing composition to the web, operating a second imager unit for applying a second printing composition to the web, and operating a closed-loop control system that controls web handling and a distance of at least one printhead of at least one of the first imager unit and the second imager unit from the web. The first printing composition comprises at least one of a polymer, a defoamer agent, a surfactant, a surface treatment agent, an opacifier, and water. The second printing composition comprises at least one of a pigment, a polymer, a co-solvent, a surfactant, and water.

(19) According to another aspect, a method of printing a web of polymeric heat-shrinkable material comprises the steps of operating a first imager unit for applying a first printing composition to the web, operating a second imager unit for applying a second printing composition to the web, and operating a closed-loop control system that controls web handling and a distance of at least one printhead of at least one of the first imager unit and the second imager unit from the web. The first printing composition comprises at least one of a viscosity modifier, a polymer, a surfactant, a defoamer agent, a surface treatment agent, an antimicrobial agent, and water. The second printing composition comprises at least one of a pigment, a polymer, a co-solvent, a surfactant, and water.

(20) According to another aspect, a printing system for printing a heat-shrinkable web comprises a transport apparatus adapted to transport a flexible web along a process direction and first and second individually controllable ink jet imager units offset from one another along the process direction wherein each of the first imager unit and the second imager unit includes a first portion operable to print on a first portion of the web and a second portion operable to print on the second portion of the web wherein each of the first portion and second portion of the first and second imager units is stationary along the process direction and the lateral direction. At least one dryer is operable to dry print on the web without substantial shrinking of the web. A first closed-loop control system is responsive to an indication of web temperature and controls the at least one dryer and a second closed-loop control system controls registration between the first imager unit and the second imager unit.

(21) According to another aspect, a printing system for printing a heat-shrinkable web comprises a transport apparatus adapted to transport a flexible web along a process direction and first and second individually controllable ink jet imager units offset from one another along the process direction. Each of the first imager unit and the second imager unit includes a first portion operable to print on a first portion of the web and a second portion operable to print on the second portion of the web wherein each of the first portion and second portion of the first and second imager units is stationary along the process direction and the lateral direction and wherein the first imager unit prints on the web at a first resolution and the second imager unit prints on the web at a second resolution different than the first resolution. At least one dryer is operable to dry print on the web without substantial shrinking of the web. A first closed-loop control system is responsive to an indication of web temperature that controls the at least one dryer. A second closed-loop control system controls registration between the first imager unit and the second imager unit wherein the second closed-loop control system is adapted to register first content printed by the first portion of the first imager unit with content printed by the first portion of the second imager unit, register content printed by the second portion of the first imager unit with content printed by the second portion of the second imager unit, independently control the first portion and the second portion of

the first imager unit, and independently control the first portion and the second portion of the second imager unit.

(22) According to yet another aspect, a printing system for printing a polymeric heat shrinkable web comprises a first imager unit for applying a first printing composition to the web and a second imager unit for applying a second printing composition to the web. The first printing composition comprises at least one of a polymer, a defoamer agent, a surfactant, a surface treatment agent, an opacifier, and water. The second printing composition comprises at least one of a pigment, a polymer, a co-solvent, a surfactant, and water.

(23) According to a still further aspect, a printing system for printing a polymeric heat shrinkable web comprises a first imager unit for applying a first printing composition to the web and a second imager unit for applying a second printing composition to the web. The first printing composition comprises at least one of a viscosity modifier, a polymer, a surfactant, a defoamer agent, a surface treatment agent, an antimicrobial agent, and water. The second printing composition comprises at least one of a pigment, a polymer, a co-solvent, a surfactant, and water.

(24) Another aspect relates to a method of printing a heat-shrinkable web comprising the steps of transporting a flexible web along a process direction and operating first and second individually controllable ink jet imager units offset from one another along the process direction to apply water-based ink to the web. Each of the first imager unit and the second imager unit includes a first portion operable to print on a first portion of the web and a second portion operable to print on the second portion of the web wherein each of the first portion and second portion of the first and second imager units is stationary along the process direction and the lateral direction. The method further includes the steps of drying print on the web without substantial shrinking of the web, controlling the at least one dryer responsive to web temperature, and controlling registration between the first imager unit and the second imager unit.

(25) According to a still further aspect, a method of printing a heat-shrinkable web comprises the steps of transporting a flexible web along a process direction and operating first and second individually controllable ink jet imager units offset from one another along the process direction to apply water-based ink to the web. Each of the first imager unit and the second imager unit includes a first portion operable to print on a first portion of the web and a second portion operable to print on the second portion of the web wherein each of the first portion and second portion of the first and second imager units is stationary along the process direction and the lateral direction. The method further includes the steps of drying print on the web without substantial shrinking of the web and controlling the at least one dryer responsive to web temperature. Still further, the method includes the step of controlling registration between the first imager unit and the second imager unit including the step of operating the second closed-loop control system to register first content printed by the first portion of the first imager unit with content printed by the first portion of the second imager unit, register content printed by the second portion of the first imager unit with content printed by the second portion of the second imager unit, independently control the first portion and the second portion of the first imager unit, and independently control the first portion and the second portion of the second imager unit.

(26) According to another aspect, a method of printing a web of polymeric heat-shrinkable material comprises the steps of operating a first imager unit for applying a first printing composition to the web and operating a second imager unit for applying a second printing composition to the web. The first printing composition comprises at least one of a polymer, a defoamer agent, a surfactant, a surface treatment agent, an opacifier, and water. The second printing composition comprises at least one of a pigment, a polymer, a co-solvent, a surfactant, and water.

(27) According to yet another aspect, a method of printing a web of polymeric heat-shrinkable material comprises the steps of operating a first imager unit for applying a first printing composition to the web and operating a second imager unit for applying a second printing composition to the web. The first printing composition comprises at least one of a viscosity

modifier, a polymer, a surfactant, a defoamer agent, a surface treatment agent, an antimicrobial agent, and water. The second printing composition comprises at least one of a pigment, a polymer, a co-solvent, a surfactant, and water.

(28) Other aspects and advantages will become apparent upon consideration of the following detailed description and the attached drawings wherein like numerals designate like structures throughout the specification.

(29) This brief description is intended only to provide a brief overview of subject matter disclosed herein according to one or more illustrative embodiments, and does not serve as a guide to interpreting the claims or to define or limit the scope of the invention, which is defined only by the appended claims. This brief description is provided to introduce an illustrative selection of concepts in a simplified form that are further described below in the detailed description. This brief description is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. The claimed subject matter is not limited to implementations that solve any or all disadvantages noted in the background.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the features of the invention can be understood, a detailed description of the invention may be had by reference to certain embodiments, some of which are illustrated in the accompanying drawings. It is to be noted, however, that the drawings illustrate only certain embodiments of this invention and are therefore not to be considered limiting of its scope, for the scope of the invention encompasses other equally effective embodiments. The drawings are not necessarily to scale, emphasis generally being placed upon illustrating the features of certain embodiments of the invention. In the drawings, like numerals are used to indicate like parts throughout the various views. Thus, for further understanding of the invention, reference can be made to the following detailed description, read in connection with the drawings in which:

(2) FIG. 1 is a simplified block diagram of an exemplary system for printing images and/or text on a substrate;

(3) FIG. 2 is an end elevational view of a polymeric film to be imaged by the system of FIG. 1;

(4) FIG. 3 is a simplified functional block diagram of the print management system of FIG. 1;

(5) FIG. 4 is a combined diagrammatic view and block diagram of an exemplary embodiment of the fourth imager unit of FIG. 1 illustrating web rotatable devices and a control system;

(6) FIG. 5 is a side elevational view of an idler roller used in the embodiment of FIG. 4;

(7) FIG. 6 is a fragmentary perspective side view of the rollers 202d and 202e of FIG. 4;

(8) FIG. 7 is a fragmentary perspective view of the idler rollers 208a, 202g, and 202f, the spreader roller 204, and the drum 72 of FIG. 4;

(9) FIG. 8 is a side elevational view of the nip roller 206 used in the embodiment of FIG. 4;

(10) FIG. 9 is an enlarged fragmentary side elevational view of a portion of the nip roller 206 of FIG. 8;

(11) FIG. 10 is fragmentary side perspective view of the spreader roller 204, nip roller 206, and drum 72 of FIG. 4;

(12) FIG. 11 is an elevational side view of the spreader roller of FIG. 10;

(13) FIG. 12 is a combined fragmentary side elevational and block view of a further portion of the imager unit and control system with associated components taken generally along the lines 12-12 of FIG. 4;

(14) FIG. 13 is a block diagram of a computer system for implementing the control system of FIG. 4;

- (15) FIG. **14** is a flowchart of programming executed by the computer system of FIG. **13**;
- (16) FIG. **15** is a flowchart of programming executed by the computer system together with hardware both as shown in FIG. **13** to implement each of the blocks **404** and **424** of FIG. **14**;
- (17) FIG. **16** is a block diagram of a computer system for implementing the print management system of FIG. **1**;
- (18) FIG. **17** is a flowchart of programming executed by the print management system of FIG. **16**;
- (19) FIG. **18** is simplified plan view of a portion of the web of FIG. **1** illustrating application of a registration mark thereon;
- (20) FIG. **19** is an enlarged fragmentary view of the registration mark of FIG. **18**;
- (21) FIG. **20** is a fragmentary plan view of the web with content portions printed in two lanes of the web;
- (22) FIG. **21** is an enlarged plan view of one of the printed content portions of FIG. **20**;
- (23) FIG. **22** is fragmentary plan view of a portion of the web with imager units and sensors;
- (24) FIG. **23** is a fragmentary plan view of the web with content portions printed in five lanes of the web;
- (25) FIGS. **24A** and **24B** illustrates the effects of shrinking a web on images printed thereon;
- (26) FIG. **25** is a block diagram of a distortion correction process of the print management system of FIG. **3**;
- (27) FIG. **26** is a flowchart of steps undertaken by the distortion corrector process of the distortion correction process of FIG. **25**;
- (28) FIG. **27** is a flowchart of steps undertaken by an on-press distortion analyzer of the distortion correction process of FIG. **25**;
- (29) FIG. **28** is a flowchart of steps undertaken by an in-plant distortion analyzer of the distortion correction process of FIG. **25**;
- (30) FIG. **29** is a flowchart of steps undertaken at a customer site to generate images of bags produced from the web printed on by the system of FIG. **1**;
- (31) FIG. **30** is a flowchart of a customer site distortion analyzer of the distortion correction process of FIG. **25**;
- (32) FIGS. **31A** and **31B** graphically illustrate determining the distortion to an image printed on a web by the system of FIG. **1**;
- (33) FIG. **32** is a simplified block diagram of a dryer unit of the system of FIG. **1**;
- (34) FIG. **32A** is a block diagram of a computer system for implementing a closed-loop dryer management system of FIG. **32**;
- (35) FIG. **33** is a flowchart of steps undertaken by a global dryer control system of the system of FIG. **1** to configure operating parameters of the dryer unit of FIG. **32**;
- (36) FIG. **34** is a flowchart of steps undertaken by a closed-loop dryer controller to control the dryer unit of FIG. **32**;
- (37) FIG. **35** is a flowchart of steps undertaken by the closed-loop dryer controller to determine if drying of material is insufficient;
- (38) FIG. **36** is a flowchart of steps undertaken by the closed-loop dryer controller to reduce a temperature of a web printed on by the system of FIG. **1**;
- (39) FIG. **37** is a flowchart of steps undertaken by the closed-loop dryer controller to raise a temperature of a web printed on by the system of FIG. **1**; and
- (40) FIGS. **38A** and **38B** are a simplified block diagram showing a temperature sensor of the dryer unit of FIG. **32**.

DETAILED DESCRIPTION

(41) FIG. **1** shows an exemplary system **20** for printing content (e.g., images and/or text) on a substrate, such as a shrinkable plastic film used in food grade applications. It should be understood, however, that the system **20** may be used to print on any polymer or other flexible material that is dimensionally stable or unstable during processing for any application, e.g., other than food grade.

The system **20** preferably operates at high-speed, e.g., on the order of zero to about 500 or more feet per minute (fpm) and even up to about 1000 fpm, although the system may be operable at a different speed, as necessary or desirable. The illustrated system **20** is capable of printing images and/or text on both sides of a substrate (i.e., the system **20** is capable of duplex printing) although this need not be the case. In the illustrated embodiment, a first side of a substrate is imaged by a sequence of particular units during a first pass, the substrate is then turned over and the other side of the substrate is imaged by all of the particular units or only by a subset of the particular units during a second pass. First portions of one or more of the particular units may be operable during the first pass and second portions of one or more of the particular units laterally offset from the first portions may be operable during the second pass. Also, one or more of the particular units may be capable of simultaneously treating and/or imaging both sides of the substrate during one pass, in which case such unit(s) need not be operable during the other pass of the substrate. In the illustrated embodiment, the first portions are equal in lateral extent to the second portions, although this is not necessarily the case. Thus, for example, the system may have a 52 inch width, and may be capable of duplex printing up to a 26 inch wide substrate. Alternatively, a 52 inch wide (or smaller) substrate may be printed on a single side (i.e., simplex printed) during a single production run. If desired, additional imager units and associated dryer and web guide units may be added in line with the disclosed imager units and other units so as to obtain full-width (i.e., 52 inch in the disclosed embodiment) duplex printing capability. Still further, a substrate having a different width, such as 64 inches (or larger or smaller width) may be accommodated.

(42) Further, the illustrated system **20** may comprise a fully digital system that solely utilizes ink jet printers, although other printing methodologies may be utilized to image one or more layers, such as flexographic printing, lithographic offset printing, silk screen printing, intaglio printing, letterpress printing, etc. Ink jet technology offers drop on demand capability, and thus, among other advantages, allows high levels of color control and image customization.

(43) In addition to the foregoing, certain ink jet heads are suitable to apply the high opacity base ink(s) that may be necessary so that other inks printed thereon can receive enough reflected white light (for example) so that the overprinted inks can adequately perform their filtering function. Some printhead technologies are more suitable for flood coating printing, like printing overcoat varnish, primers, and white, and metallic inks.

(44) On the other hand, printing high fidelity images with high resolution printheads achieves the best quality. Using drum technology and printing with ink jet is the preferred way to maintain registration, control a flexible/shrinkable film substrate, and reproduce an extended gamut color pallet.

(45) The system disclosed herein has the capability to print an extended gamut image. In some cases the color reproduction required may need a custom spot color to match the color exactly. In these cases, an extra eighth channel (and additional channels, if required) can be used to print custom color(s) in synchronization with the other processes in the system.

(46) Printing on flexible/shrinkable films with water-based inks has many challenges and require fluid management, temperature control, and closed loop processes. Thus, in the present system, for example, the ability to maintain a high quality color gamut at high speed is further process controlled by sensor(s) that may comprise one or more calibration cameras to fine tune the system continually over the length of large runs.

(47) As used herein, the phrase “heat-shrinkable” is used with reference to films which exhibit a total free shrink (i.e., the sum of the free shrink in both the machine and transverse directions) of at least 10% at 185° F., as measured by ASTM D2732, which is hereby incorporated, in its entirety, by reference thereto. All films exhibiting a total free shrink of less than 10% at 185° F. are herein designated as being non-heat-shrinkable. The heat-shrinkable film can have a total free shrink at 185° F. of at least 15%, or at least 20%, or at least 30%, or at least 40%, or at least 45%, or at least 50%, or at least 55%, or at least 60%, or at least 65%, or at least 70%, as measured by ASTM

D2732. Heat shrinkability can be achieved by carrying out orientation in the solid state (i.e., at a temperature below the glass transition temperature of the polymer). The total orientation factor employed (i.e., stretching in the transverse direction and drawing in the machine direction) can be any desired factor, such as at least 2×, at least 3×, at least 4×, at least 5×, at least 6×, at least 7×, at least 8×, at least 9×, at least 10×, at least 16×, or from 1.5× to 20×, from 2× to 16×, from 3× to 12×, or from 4× to 9×.

(48) As shown in FIG. 1, the illustrated system **20** includes a first pull module **22** that unwinds a web of plastic web **24** from a roll **25** that is engaged by a nip roller **23** at the beginning of a first printing pass through the system **20**. The web **24** may comprise a flattened cylinder or tube of plastic film comprising two layers having sides **24a**, **24b** (see FIG. 2) joined at side folds **24c**, **24d**, although the web **24** may instead simply comprise a single layer of material, if desired and as referred to above. Once unwound by the module **22**, the web **24** may be processed by a surface energy modification system, such as a corona treatment unit **26** of conventional type, that increases the surface energy of the web **24**. The corona treatment addresses an imaging condition that may be encountered when a large number of closely spaced drops are applied to a low surface energy impermeable material, which, if not compensated for, can result in positional distortion of the applied inks due to coalescence effects. The corona treatment module may be capable of treating both sides of the web **24** simultaneously. A first web guide **28** of conventional type that controls the lateral position of the web **24** in a closed-loop manner then guides the corona-treated web **24** a first imager unit **30**. A first dryer unit **32** is operated to dry the material that is applied to the web **24** by the first imager unit **30**. The material applied by the first imager unit **30** may be deposited over the entirety of the web **24** or may be selectively applied only to some or all areas that will later receive ink.

(49) A second pull module **40** and a second web guide **42** (wherein the latter may be identical to the first web guide **28**) deliver the web **24** to a second imager unit **44** that prints a material supplied by a first supply unit **45** on the web **24**. A second dryer unit **46** is operable to dry the material applied by the second imager unit **44**.

(50) Thereafter, the web **24** is guided by a third web guide **48** (again, which may be identical to the first web guide **28**) to a third imager unit **60** that applies material supplied by a second supply unit **62** thereon, such as at a location at least partially covering the material that was deposited by the second imager unit **44**. A third dryer unit **64** is operable to dry the material applied by the third imager unit **60** and the web **24** is then guided by a fourth web guide **66** (that also may be identical to the first web guide **28**) to a fourth imager unit **70** comprising a relatively high resolution, extended color gamut imager unit **70**.

(51) The imager unit **70** includes a drum **72** around which are arranged ink jet printheads for applying primary process color inks CMYK to the web **24** along with secondary process color inks orange, violet, and green OVG and an optional spot color ink S to the web **24** at a relatively high resolution, such as 1200 dpi and at a high speed (e.g., 100-500 fpm). The extended gamut printing is calibrated at the high printing speed. The drop sizes thus applied are relatively small (on the order of 3-6 pL). If desired, the imager unit **70** may operate at a different resolution and/or apply different drop sizes. The inks are supplied by third and fourth supply units **74**, **76**, respectively, and, in some embodiments, the inks are of the water-based type. The process colors comprising the CMYK and OVG inks enable reproduction of extended gamut detailed images and high quality graphics on the web **24**. A fourth dryer unit **80** is disposed downstream of the fourth imager unit **70** and dries the inks applied thereby.

(52) Following imaging, the web **24** may be guided by a web guide **81** (preferably identical to the first web guide **28**) and coated by a fifth imager unit **82** comprising an ink jet printer operating at a relatively low resolution and large drop size (e.g., 600 dots-per-inch (dpi), 5-12 picoliter (pL) size drops) to apply an overcoat, such as varnish, to the imaged portions of the web **24**. The overcoat is dried by a fifth dryer unit **84**. Thereafter, the web is guided by a web guide **88** (also preferably

identical to the first web guide **28**), turned over by a web turn bar **90**, which may comprise a known air bar, and returned to the first pull module **22** to initiate a second pass through the system **20**, following which material deposition/imaging on the second side of the web **24** may be undertaken, for example, as described above. The fully imaged web **24** is then stored on a take-up roll **100** engaged by a nip roll **101** and thereafter may be further processed, for example, to create shrink-wrap bags.

(53) While the web **24** is shown in FIG. **1** as being returned to first the pull module **22** at the initiation of the second pass, it may be noted that the web may be instead delivered to another point in the system **20**, such as the web guide **28**, the first imager unit **30**, the pull module **40**, the web guide **42**, or the imager unit **44** (e.g., when the web **24** is not to be pre-coated), bypassing front end units and/or modules, such as the module **22** and the corona treatment unit **26**.

(54) Further, in the case that the web **24** is to be simplex printed (i.e., on only one side) the printed web **24** may be stored on the take-up roll **100** immediately following the first pass through the system **20**, thereby omitting the second pass entirely.

(55) The web **24** may be multilayer and may have a thickness of 0.25 mm or less, or a thickness of from 0.5 to 30 mils, or from 0.5 to 15 mils, or from 1 to 10 mils, or from 1 to 8 mils, or from 1.1 to 7 mils, or from 1.2 to 6 mils, or from 1.3 to 5 mils, or from 1.5 to 4 mils, or from 1.6 to 3.5 mils, or from 1.8 to 3.3 mils, or from 2 to 3 mils, or from 1.5 to 4 mils, or from 0.5 to 1.5 mils, or from 1 to 1.5 mils, or from 0.7 to 1.3 mils, or from 0.8 to 1.2 mils, or from 0.9 to 1.1 mils. The web **24** may have a film percent transparency (also referred to herein as film clarity) measured in accordance with ASTM D 1746-97 "Standard Test Method for Transparency of Plastic Sheeting", published April, 1998, which is hereby incorporated, in its entirety, of at least 15 percent, or at least 20 percent, or at least 25 percent, or at least 30 percent.

(56) Preferably, the system **20** includes a first tension zone between the roll **25** (which is a driven roll) and the pull module **22**, a second tension zone between the pull module **22** and the imager unit **30**, a third tension unit between the imager unit **30** and the pull module **40**, a fourth tension zone between the pull module **40** and the imager unit **44**, a fifth tension zone between the imager unit **44** and the imager unit **60**, a sixth tension zone between the imager unit **60** and the drum **72**, a seventh tension zone between the drum **72** and the imager unit **82**, and an eighth tension zone between the imager unit **82** and the take-up roll **100** (which is a driven roll). One or more tension zones may be disposed between the imager unit **82** and the pull module **22** and/or at other points in the system **20**. Each of the elements defining the ends of the tension zones comprises, for example, a driven roll (which, in the case of the imager units **30**, **44**, **60**, **70**, and **82**, comprise imager drums) with a nip roller as described in greater detail hereinafter. Preferably, all of the tension zones are limited to about 20 feet or less in length. The web tension in each tension zone is controlled by one or more tension controllers such that the web tension does not fall outside of predetermined range(s).

(57) The nature and design of the first, second, and third imager units **30**, may vary with the printing methodologies that are to be used in the system **20**. For example, in a particular embodiment in which a combination of flexographic and ink jet reproduction is used, then the first imager unit **30** may apply a composition comprising a clear primer and a dispersion of a white colorant, such as titanium dioxide, in a flood-coated fashion to the web **24**. The second imager unit **44**, which may comprise an ink jet printer or a flexographic unit, may thereafter deposit one or more metallic ink(s) onto the web at least in portions that received material from the first imager unit **30**. In such an embodiment, the third imager unit **60** is not required, and the imager unit **60** and dryer unit **64** and web guide **66** associated therewith may be omitted.

(58) In a further embodiment, the first imager unit **30** comprises a flexographic unit that applies a white pigmented ink to the web **24**, the second imager unit **44** comprises an ink jet printer or a flexographic unit that applies one or more metallic inks, and the third imager unit **60** comprises an ink jet printer or flexographic unit that applies a clear primer to the web **24**.

(59) In yet another embodiment that uses ink jet technology throughout the system **20**, the first

imager unit **30** comprising an ink jet printer may apply a composition comprising a clear primer and a dispersion of a white colorant, such as titanium dioxide, to the web **24**. The second imager unit **44**, which comprises an ink jet printer, may thereafter deposit one or more metallic ink(s) onto the web at least in portions that received material from the first imager unit **30**. In such an embodiment, the third imager unit **60** is not required, and the imager unit **60** and dryer unit **64** and web guide **66** associated therewith may be omitted.

(60) In a still further embodiment, the first imager unit **30** comprises an ink jet printer that applies a white pigmented ink to the web **24**, the second imager unit **44** comprises an ink jet printer that applies one or more metallic inks, and the third imager unit **60** comprises an ink jet printer that applies a clear primer to the web **24**.

(61) Any one or more of the imager units **30**, **44**, **60**, **70**, and **82** may be omitted or the functionality thereof may be combined with one or more other imager units. Thus, for example, in the case where a combined primer and white pigmented material are applied, the combination may be printed by one of the imager units **30** or **44** and the other of the imager units **30**, **44** may be omitted.

(62) In some embodiments each of the first, second, and third imager units **30**, **44**, **60** comprises a 600 dpi inkjet printer that applies relatively large drops (i.e., at least 5-12 pL) each using piezoelectric ink jet heads, although the imager units **30**, **44**, and/or **60** may operate at a different resolution and/or apply different sizes of drops. Thus, for example, a printhead designed for use with metallic and precoating inks in the present system may have a resolution of 400 dpi and drop volume of 20-30 pL. The pre-coating material, white, and metallic inks have relatively heavy pigment loading and/or large particle sizes that are best applied by the relatively low resolution/large drop size heads of the imager units **30**, **44**, **60**.

(63) In alternative embodiments, one or more of the primer, white, and coating imager units may operate at a relatively high resolution and/or small drop size, such as 1200 dpi/3-6 pL.

(64) The primer renders at least a portion of the surface of the web **24** suitable to receive later-applied water-based inks. It is preferable (although not necessary) to apply the primer just before the process and spot color inks are applied by the fourth imager unit **70** so that the such colors are directly applied to the dried primer.

(65) Preferably, the fourth imager unit **70** comprises the above-described ink jet printer so that drop-on-demand technology may be taken advantage of, particularly with respect to print-to-print variability, high resolution, and the ability to control registration precisely.

(66) The fifth imager unit **82** also preferably comprises an ink jet printer that operates at least at 1200 dpi or 2400 dpi, although it may instead be implemented by a different printing methodology, such as a flexographic unit.

(67) As noted in greater detail hereinafter, a supervisory or global control system **120** is responsive to sensors (not shown in FIG. **1**) and is responsible for overall closed-loop control of various system devices during a production run. A further control system comprising a print management control system **130** controls the various imager units also in a closed-loop fashion to control image reproduction as well as color correction, registration, correct for missing pixels, etc. The print management control system **130** controls the various imager units **30**, **44**, **60**, **70**, and **82**. For example, the imager unit **70** includes first and second imager portions **225**, **227** wherein each imager portion **225**, **227** comprises one or two printheads for each of the colors CMYK and OVG and the spot color S for a total of sixteen printheads (in the case that there is a single printhead per imager portion). Eight of the printheads **226a-226d** and **228a-228d**.

(68) Also in the illustrated embodiment, each dryer unit **32**, **46**, **64**, **80**, and **84** is controlled by an associated closed-loop dryer management system (not shown in FIG. **1**) during printing to, among other things, minimize image offsetting (sometimes referred to as "pick-off"), which can result in artifacts that may result from improper or insufficient drying of ink deposited on the web causing undried ink/coating to adhere (i.e., offset) to one or more system handling components, such as idler roller(s) or other component(s), and be transferred from such system handling component(s)

to other portions of the web.

(69) In the case of a partially or completely ink jet implemented system, the printheads used by the first through fifth imager units **30**, **44**, **60**, **70**, and/or **82** may be of the same or different types, even within each printer, and/or, as noted previously, different printing methodologies could be used to apply inks/coatings. In any event, the global control system **120** and/or the print management control system **130** is (are) programmed to convert input data representing the various layers, such as data in a print-ready source format (e.g., Adobe Portable Document Format or PDF) to bitmaps by a ripping process or other page representation(s) during pre-processing taking into account the operational characteristics of the various printhead types/printing methodologies (such as the resolution(s) and drop size(s) to be deposited) and properties of the web (such as shrinkage when exposed to heat).

(70) In addition to the foregoing, one or more additional control systems may be provided, for example, to track and control the web **24** as the web **24** is conveyed through the system **20** and as described further hereinafter. The various control systems may be implemented together or separately by one or more suitable programmable devices, input sensors, and output control devices, as appropriate or desirable.

(71) Referring next to FIG. **3**, an exemplary embodiment of the print management control system **130** is illustrated in generalized form, it being assumed that the first imager unit **30** applies pre-coating material over a selected portion of or over the entire web **24** so that control of such imager unit **30** is straightforward and therefore not illustrated. The exemplary print management control system **130** takes in pages **150** in a print-ready format, such as PDF or another print-ready or non-print-ready format, and divides each page into data representing layers that are to be imaged by the imager units **44**, **60**, **70**, and **82**. More particularly, using the illustrated page **150** as an example, a processing unit **152** divides the data defining the page **150** into layer data representing four layers **150a**, **150b**, **150c**, and **150d** to be printed in white, silver, process colors (with an optional spot color), and overcoat, respectively, color corrects the layer data as needed taking into account the particular inks and web material, and converts the color corrected layer data into four layer bitmaps using a raster image processing (RIP) technique (block **154**). The processing unit **152** then determines registration parameters that are used in conjunction with the layer bitmaps to control the individual imager units **44**, **60**, **70**, and **82** (block **156**) such that the layer images are accurately printed atop one another on the web **24**.

(72) The processing unit **152**, which may comprise a suitably programmed computer or server or other programmable device, is responsive to feedback signals developed by sensors including a position encoder **160** and, optionally, a camera **162** that sense web position and the printed image so that the processing unit **152** and/or other controls can operate in a closed-loop manner during start up, shutdown and steady state operation.

(73) It has been found that digitally imaging heat shrinkable extensible tube material presents web handling issues due to the risk of printhead damage from wrinkles and splices that are not a risk for normal flexographic imaging processes. Wrinkles in extensible film webs can be formed in several ways: 1.) air trapped in the web **24** forms pockets due to smooth nip points and the pockets wrap over solid idler rollers that inadvertently burst the air pockets during web movement and deform the material surface; 2.) the distance between contact points may be excessive, thereby allowing the material to fold onto itself; 3.) the alignment tolerances between contact rollers may be inadequately controlled, leading to wrinkle formation; and 4.) standard tension control methods are typically not sufficiently precisely controllable to avoid wrinkling.

(74) In order to address these issues a web handling system **220**, a portion of which is shown in FIG. **4**, manages the travel of the web **24** to and from the fourth imager unit **70**, for example. (It should be noted that the fourth imager unit **70** is inverted front-to-back as compared to the showing thereof in FIG. **3**). Similar and/or identical components may be used to control the movement of the web at other portions of the system **20**, as described in greater detail hereinafter. The web

handling system **220** comprises journaled infeed idler rollers **202a-202g**, a journaled spreader roller **204**, a journaled nip roller **206** disposed adjacent the drum **72**, and journaled outfeed idler rollers **208a-208d**. It should be noted that a greater or lesser number of rollers may instead be used to transport the web **24**, as necessary or desirable.

(75) Referring to FIGS. **5**, **6**, and **10**, in the illustrated embodiment the idler rollers **202** and **208** are identical to one another and each of the idler rollers **202**, **208** is fabricated of a metal or other material. Referring specifically to FIG. **5**, the idler roller **202a** comprises a cylindrical surface **202a-1** and diagonally-extending grooves **202a-2** and **202a-3** that cross one another, preferably, but not necessarily, at right angles. The grooves **202a-2** and **202a-3** are all identical to one another. Each of the idler rollers **202**, **208**, such as the idler roller **202a**, preferably comprises two independently journaled (i.e., split) portions comprising halves **202a-4** and **202a-5** that are separated by a small distance, such as one-ten thousandths of an inch, so that the halves **202a-4** and **202a-5** can rotate in response to the passage of, for example, a 52 inch web thereover, or can rotate independently at different speeds in response to the passage of, for example, two 26 inch webs thereover. Preferably, each of the grooves **202a-2** and **202a-3** of each portion **202a-4** and **202a-5** has a V-shape or a U-shape that extends continuously from one axial end of each roller portion **202a-4** and **202a-5** to the other axial end of the roller portion **202a-4** and **202a-5**. The spreader roller **204** may comprise any known or conventional spreader roller of any suitable type, such as a fixed bow roll, an adjustable bow roll, a concave web spreading roll, an ESR segmented expander roll, or an expander web spreading roll.

(76) In the illustrated embodiment, the spreader roller **204** comprises a conventional resilient cylindrical roller with two spiral grooves **204a** and **204b** (FIG. **11**).

(77) Referring to FIGS. **7-9**, the nip roller **206** is also of conventional or known design and comprises a resilient outer surface **206a** and a plurality of grooves **206b**. The grooves **206b** are disposed perpendicular to a longitudinal axis of the roller **206** and are therefore parallel to one another. In the illustrated embodiment the grooves **206b** have identical dimensions to one another and are equally spaced along the roller **206**, although some or all of the grooves **206b** may have different dimensions than some or all of the remaining grooves **206b**. Further, in the illustrated embodiment each of the grooves **206b** has a rectangular cross-sectional shape comprising a width parallel to the longitudinal axis of one-sixteenth of an inch, a depth of one-sixteenth of an inch, and a spacing between centers of adjacent grooves of one-quarter inch. Further, the outer surface **206a** may be made of rubber or other suitable material. During operation, air trapped in the web **24** collects in the grooves **206b** and passes through the nip with the drum **72** so that the air is not allowed to accumulate behind the nip and possibly stretch or burst the web **24**.

(78) Preferably, each idler roller **202** and **208** is spaced center to center from adjacent rollers **202**, **204**, **206**, and **208** in a range between 38 to 28 inches, more preferably in a range between 36 to 30 inches, and most preferably in a range between 35 to 33 inches. Alternatively, each idler roller **202** and **208** is spaced center to center from adjacent rollers **202**, **204**, **206**, and **208** no more than about 36 inches, and more preferably no more than about 34 inches, and most preferably less than or equal to about 30 inches so that the unsupported length of the web **24** is limited at all points in the system **220**. Also, roller-to-roller alignment is precisely controlled by ensuring that the centerline of every roller **202**, **204**, **206**, and **208** in the system is preferably aligned no more than about 0.001 inches per foot along the longitudinal axis of each roller to a selected single virtual datum point, and more preferably no more than about 0.00075 inches per foot along the longitudinal axis of each roller to a selected single virtual datum point, and most preferably less than or equal to about 0.0005 inches per foot along the longitudinal axis of each roller to a selected single virtual datum point. In addition, a spreader roller, such as the roller **204**, is disposed at no greater than a particular distance before every critical nip point in the system **20**. Specifically, the spreader roller **204** is preferably disposed a distance from the nip point between the nip roller **206** and the drum **72** in a range between about 6 inches to about 11 inches, more preferably in a range between about 6.5

inches and about 9.0 inches, and most preferably between about 7.0 inches and about 8.5 inches as measured between the point at which the web **24** leaves contact with the spreader roller **204** and the nip point. Alternatively, the spreader roller **204** is preferably disposed a distance from the nip point between the nip roller **206** and the drum **72** about 10 inches or less, more preferably about 8.5 inches or less, and most preferably about 7 inches or less as measured between the point at which the web **24** leaves contact with the spreader roller **204** and the nip point to maintain wrinkle free material in the nip. Thus, the system **220** may have the foregoing parameter magnitudes comprising a roller spacing between adjacent rollers of no greater than 34 inches, an alignment no greater than about 0.001 inches per foot along the longitudinal axis of each roller to a selected single virtual datum point, and a distance of no greater than about 7 inches between a spreader roller and a nip point. One might alternatively use any other combination(s) of the foregoing recited parameter magnitudes as desired, such as a roller spacing of about 36 or about 30 inches, an alignment of about 0.00075 or about 0.0005 inches per foot, and a distance of about 10 inches or about 8.5 inches between a spreader roller and a nip point.

(79) Each element defining the ends of the tension zones comprises a nip roller as seen in FIGS. **8** and **9** adjacent a driven roll or drum. Further, with the exception of the roll **25**, a spreader roller such as the one shown in FIG. **11** and/or as described above is disposed upstream of each nip at the ends of the tension zones. Also, the web **24** is supported at the spacings described above within each tension zone by idler rollers similar or identical to the rollers **202**, **208**.

(80) The system **200** may also incorporate a printhead gap control system. Further, while the foregoing is effective to minimize the incidence of wrinkle formation, wrinkling might still occur and/or splices may need to be accommodated. Thus, provision is made as described below to control printhead gapping and prevent damage to one or more ink jet printheads. While the control system **222** described below is shown in connection with the imager unit **70**, identical or substantially similar control systems are used in connection with the remaining imager units **30**, **44**, **60**, and/or **82**, as should be evident. If desired, elements of the various control systems may be combined and/or shared and/or the systems may be completely separate. Inasmuch as the control system **222** controls the position of sixteen printheads, and the remaining imager units use fewer heads and operate potentially at different resolutions and/or drop sizes, the control systems other than that described hereinafter must be modified to take these differences into account, as should be evident to one of ordinary skill in the art. In the illustrated embodiment, first portions of the imager units **30**, **44**, **60**, **70**, and **82** print on the first side **24a** of the web **24** and second portions of the imager units **30**, **44**, **60**, **70**, and **82** print on the second side **24b** of the web **24**.

(81) As seen in FIGS. **4** and **12**, a plurality of thickness sensors **223** of any suitable type senses one or more thicknesses of the web **24**, for example, at spaced points along the length or width thereof. The multiple sensors **223** may instead be replaced by a single sensor, such as a CCD camera that extends across the full or partial width of the web **24**, if desired. The control system **222** is responsive to the output(s) of the sensors **223** and comprises and controls a plurality of actuators **224** that control the distance of the faces of various printheads of two imager portions each comprising one or two printheads, wherein a printhead **226** and **228** of each portion are shown in generalized form in FIG. **4**, eight of which **228a-228h** are shown in FIG. **3**, and eight of which **226a-226d** and **228a-228d** are shown in FIG. **12**, it being understood that there are sixteen printheads in total comprising two for each of the colors CMYK and OVG and the spot color S (in the event that each imager portion includes a single printhead). Specifically, with reference to FIG. **12**, the printheads **226a**, **228a** are independently operable and disposed in side-by-side relationship to apply cyan up to the full width of the web **24**, the printheads **226b**, **228b** are disposed in side-by-side relationship and are independently operable to apply magenta up to the full width of the web **24**, and so on for the remaining printheads (as seen in FIG. **3** the printheads **226**, **228** are disposed about the periphery of the drum **72** and the printheads **226**, **228** for the colors OVGs are disposed behind the drum **72** of FIG. **12** and are thus not visible in such FIG.). The printhead **226** for each

color is laterally directly adjacent the printhead **228** for the same color (i.e., the innermost ejection orifices or ports of the printheads **226**, **228** are spaced substantially equal to the spacing between the remaining adjacent orifices or ports of the printheads **226**, **228**) so that a full-width web may be imaged without creating a lateral gap between the portions imaged by the printheads **226**, **228** on the web **24**. Further, each of the printheads **226**, **228** (as well as each of the printheads in the imager units **30**, **44**, **60**, and **82**) is stationary in the process direction and the lateral direction and is radially movable, preferably independently from one another, toward and away from the drum **72**, and thus, from the web **24**, by the actuators **224** responsive to the sensors **223** and remainder of the control system **222**. The positions of the printheads **226**, **228** may be determined by sensors, such as the position sensor **229** for one or the printheads **226** (like sensors are provided to sense the positions of the remaining printheads) and the actuators **224** may be controlled in a closed-loop fashion to obtain precise positioning. The system **222** thus allows for dynamic closed-loop printhead gapping from each printhead face to the drum **72** depending on the substrate thickness based on feedback developed by the sensors **223**. In this regard, a web position encoder **230**, which may be an optical device, (and which may comprise the position encoder **160** or may be separate therefrom) senses the web position and/or speed, for example by detecting sense marks printed on the web **24**, during movement thereof so that the printheads **226** and/or **228** are properly positioned for optimal imaging as the web thickness changes at the drum **72**. If desired, the gapping of one printhead may be the same as or different than the gapping of other printhead(s). In the preferred embodiment a printhead gapping distance of about 0.0405 to about 0.052 inches for substrate thicknesses ranging from about 0.0005 to about 0.012 inches can be accommodated, although a different gapping range, and hence, substrate thickness range, might alternatively be accommodated.

(82) The thickness sensors **223** are also capable of detecting a splice and/or wrinkles in the web **24**. Alternatively, a dedicated splice/wrinkle sensor **231** (FIGS. **4** and **13**) of a conventional optical or other suitable variety may be provided at any suitable point of the web travel. For example, one may sense opacity increases or ultrasonic signal attenuation from multiple layers or splicing tape. In the event that a splice or wrinkle is detected, the control system **222**, in response to a signal from the sensors **223** or **231**, senses the output of the web position encoder **230** and, at the appropriate time just before the splice or wrinkle reaches the drum **72**, temporarily retracts all of the printheads **226**, **228** so that the splice or wrinkle does not damage any of the printheads. The control system **222** moves the printheads **226**, **228** back to their appropriate gapping distances once the splice or wrinkle has passed the printheads.

(83) The control system **222** also comprises a tension control that is responsive to one or more strain gauges disposed in one or more of the rollers **202**, **208** (such as a strain gauge **202c-1** in the roller **202c** of FIGS. **4** and **13**) and/or other rollers in other tension zones and controls the speed of one or more driven rollers in the system **20**, such as a drive motor **73** (FIG. **13**) that controls the movement of the drum **72** and a drive motor **233** in the third imager unit **60** that supplies motive power to a driven roller **235**, to control tension in the web **24** at each tension zone, such as the sixth tension zone.

(84) FIG. **13** illustrates a computer system **300** especially adapted to implement the control system **222**, it being understood that any or all of the control systems disclosed herein, such as one or more of the control systems **120**, **130**, and/or the dryer control system(s), may be implemented by like computer systems or by the computer system **300**. Thus, for example, the computer system **300** may also comprise the processing unit **152** and may implement the control system **222**. The computer system **300** comprises a personal computer, server, or other programmable device **302** having a memory **304** that, among other things, stores programming executed by a processing module or controller **306** to implement the control system **222**. The device **302** receives signals from the strain gauges including the strain gauge **202c-1**, the web position encoder **230**, and the sensors **223**, **229**, and **231** and controls the actuators **224** and the various drive motors, such as the

drive motor **73** and drive motor **233** as noted below.

(85) The programming illustrated in FIG. **14** is executed by the device **302** to implement the control system **222**. The programming begins at a block **350** that detects when an operator has pushed a “jog” button **352** (FIG. **13**) after first preloading the system **20** with the material of the web **24**. In the latter regard, the web **24** is preferably loaded only through those system components that are to be active, and therefore enabled, during the pass(es) through the system **20**, thereby bypassing unused system components.

(86) Once the block **350** determines that the operator has pressed the jog button **352**, control passes to a series of blocks that execute a pre-tensioning and web characterization sequence. A block **356** commands a driven roller in the first tension zone to eliminate slack in the first tension zone. Referring again also to FIG. **1**, assuming that all of the components of the system **20** are enabled for use, the block **356** commands the driven roller in the pull module **22** to rotate until a target tension in the first tension zone is achieved, at which point the driven roller is maintained at such position. A block **358** then measures the stretch in the first tension zone so as to obtain a characterization of the substrate in the first tension zone. The stretch is calculated by the block **358** (and by subsequent blocks) using roller encoders disposed in idler rollers, such as a roller encoder **359** in the roller **202c** of FIGS. **4** and **13**, together with the tension sensed by one or more strain gauges in the respective tension zone, wherein the strain gauge(s) may be similar or identical to the strain gauge **202c-1** described above.

(87) A block **360** identical to the block **356** commands the driven roller in the second tension zone comprising the driven roller in the first imager unit **30** to rotate and remove slack in the second tension zone until a target tension for the second tension zone is achieved. A block **362** identical to the block **358** then measures the stretching of the web **24** in the second tension zone so as to characterize the portion of the web **24** in such zone. Subsequent blocks identical to the blocks **356** and **358**, such as blocks **370**, **372**, sequentially remove slack in the third through eighth remaining tension zones, tension the web **24** to target values, and measure stretch in each of the zones to characterize the web **24** in each of the third through eighth tension zones.

(88) Following the block **372**, a block **380** receives data concerning the substrate web **24** and calculates the modulus of elasticity of the material of the web **24**. The data, which may be supplied by the operator, another person, or by indicia, such as a sensed barcode, may comprise information concerning the material of the web **24**. The modulus of elasticity, the web characterization undertaken by the blocks **358**, **362**, **372** and corresponding blocks for other web tension zones, or both, may be used at a subsequent point in the programming as noted in greater detail hereinafter to establish PID controller parameters. Also, if the web characterization undertaken by the blocks **358**, **362**, **372** and corresponding blocks for other web tension zones indicates that there is a significant discrepancy between the measured characterization and the substrate data and/or the calculated substrate modulus of elasticity, action may be undertaken, such as immediately disabling the system **20**, energizing a light and/or audible alarm, etc.

(89) Also following the block **372** control pauses until the operator again presses the jog button **352** whereupon execution passes to a block **382** that initiates a production run. The block **382** supplies electrical power to the various motors and associated motor drives, such as the motor drives **384**, **386** of FIG. **13**.

(90) Following the block **382**, a block **394** releases a rewind brake **396** (FIG. **13**) associated with the take-up roll **100** (FIG. **1**). A block **397** thereafter resets proportional-integral-differential (PID) controllers **398** (FIG. **13**) two of which are associated with one of the enabled tension zones. The PID controllers are implemented by the device **302**. A block **400** then sets proportional and integral gains for each PID controller **398** to predetermined values.

(91) Following the block **400**, a block **402** checks to determine whether the web **24** is to be simplex printed or duplex printed. If the web is to be printed only on one side, control passes to a block **404** described in greater detail hereinafter that operates the various driven rollers for the enabled units

so that a commanded system throughput is achieved. A block **408** checks to determine whether the operator has commanded that the system **20** be stopped. If not, control returns to the block **404**, and control remains with the block **404** until the operator has commanded that the system **200** be stopped, whereupon control passes to a block **410**.

(92) The block **410** engages the rewind brake **396**, a block **412** then slowly and in a controlled fashion reduces the speed commands for the various driven rollers in the enabled units, and a block **414** powers off the various motors to bring the web **24** to a controlled stop.

(93) If the block **402** determines that the web is to be printed on both sides, control passes to a block **420** that sums multiple moments of inertia about a lateral centerline of a roller, such as an idler roller **202** or **208**, in order to obtain an indication of the total tension developed by both webs in the tension zone in which the roller is disposed. The block **420** further sums multiple moments of inertia about a lateral centerline of a first one of the web portions supported by the roller **202** or **208** and the tension developed by a second one of the web portions is obtained by setting the latter sum of the moments calculated by the block **420** equal to zero in the determination of total tension.

(94) A block **422** then calculates the tension in the first web by summing the moments of inertia about a lateral centerline of the second web and setting such summed moments to zero in the determination of total tension. A block **424** that is preferably identical to the block **404** operates the various driven rollers for the enabled units at proper speeds for a commanded throughput while also controlling tension in the tension zones. A block **408** then checks to determine whether the operator has commanded that the system **20** be stopped. If not, control returns to the block **420**, and control remains in the loop comprising the blocks **420**, **422**, and **424** until the operator has commanded that the system **200** be stopped, whereupon control passes to the blocks **410**, **412**, and **414** so that the system **20** is brought to a controlled stop.

(95) FIG. **15** illustrates a combination of programming and hardware to implement each of the blocks **404** and **424** of FIG. **14**. The programming is responsive to a throughput speed command entered by an operator (FIG. **13**) to the computer system **300**. The programming includes execution branches **450a**, **450b**, . . . , **450N** that are preferably identical or similar to one another. The branch **450a** controls, for example, the motor **73**, the branch **450b** controls a motor **452** that provides motive power to a driven roller in another tension zone, such as a driven roller in the imager unit **82** disposed in the seventh tension zone. The branch **450 N** may control the drive motor **233** in the third imager unit **60**. Other driven rollers are controlled by identical or similar execution branches **450**.

(96) Inasmuch as the execution branches **450** are identical or similar, only the execution branch **450a** will be described in detail. The branch **450a** begins at a block **460a** that adjusts the throughput speed command by a first ratio that takes into account the diameter of the drum **72** so that the surface of the drum **72** moves at a commanded tangential speed to control web tension and system throughput. Next, a block **462a** further modifies the speed command by a second ratio based upon a tension feedback signal developed by a tension sensor **464a**, which may comprise one or more of the strain gauges such as the strain gauge **202c-1** (FIGS. **4** and **13**) disposed in one or more of the rollers **202**, **208**, in this case, of the sixth tension zone, wherein the tension feedback signal is modified by a first one **398a-1** of the PID controllers **398**. The resulting command signal is supplied to the motor drive **384** to operate the motor **73**. A second one **398a-2** of the PID controllers **398** is responsive to a motor position feedback signal developed by a motor position sensor **462c** and provides a modified feedback signal to the motor drive **384** so that the latter operates as a closed-loop controller. Significantly, the PID controller **398a-1** is a relatively slow controller so that tension is controlled over a relatively wide range by adjusting driven roller positions slowly.

(97) On the other hand, the PID controller **398a-2** is a relatively fast-acting controller that maintains synchronized operation of the driven rollers.

(98) FIG. **16** illustrates a computer system **300** especially adapted to implement the print

management control system **130** in a digital fashion, it being understood that any or all of the control systems disclosed herein, such as one or more of the control system **120** and/or the dryer control system(s), may be implemented by like computer systems or by the computer system **300**. Thus, for example, the system **300** may comprise the processing unit **152** and, if desired, may implement the control system **120**. The computer system **300** comprises a personal computer, server, or other programmable device **302** having a memory **304** that, among other things, stores programming as seen in FIG. **17** that is executed by a processing module or controller **306** to implement the print management control system **130**. The device **302** receives signals from various sensors, which may comprise cameras and/or other devices. Specifically, in the illustrated embodiment the device **302** is responsive to one or more image sensors, such as cameras **500**, **502** located upstream from the imager unit **70** and a further image sensor **504**, which may comprise a camera or a conventional sense mark device, which is adapted to sense a registration mark through the back side **24b** of the web **24**. The device **302** may also be responsive to a web position signal developed by the position encoder **160** and, optionally, the camera **162**. The camera **162**, when used, images the entire width of the web **24** (54 inches in the illustrated embodiment) and allows the print management control system **130** (or any of the other control systems of the system **20**) to stitch together images printed by printheads, undertake color-to-color registration and color calibration, detect missing pixel(s), and undertake printhead normalization across the web.

(99) The device **302** is also responsive to other cameras (not shown) each located upstream of other imager units **30**, **44**, **60**, and **82** and includes one or more pixel buffers **307** that store data to control the first through fifth imager units in the manner described below in connection with the fourth imager unit **70**.

(100) As is conventional, a repeating series of content portions separated by blank areas are printed along the length of the web **24**. Each content portion may comprise an image, text, or both. Thus, for example, in the illustrated embodiment of FIG. **20**, the web **24** is to be printed on the first side **24a** in two laterally-spaced lanes **556**, **558** with repeating sets of images **560**, **562** wherein the images **560**, **562** are offset along the process direction perpendicular to the lateral direction so that the content portions are separated by blank areas (only one set of images **560**, **562** is illustrated in FIG. **20**, it being understood that other equally-spaced (or non-equally-spaced) sets are printed on and along the web **24** in the process direction). It should be noted that the web **24** may be printed simplex or duplex in a different number of lane(s) and that printed content may or may not be offset relative to one another along the process direction. Also in the illustrated embodiment, the images **560**, **562** are identical, or substantially so, although the system **20** may print image(s) and/or text comprising printed content of any kind and the printed content in the lanes may be substantially or completely different.

(101) As seen in FIG. **21** each printed content portion, such as the content portion **560**, has an X-direction along the lateral direction and a Y-direction along the process direction. In the illustrated embodiment each content portion has an X-direction equal to the Y-direction wherein both are *n* units (such as inches) in width and length, respectively. Also, an origin point **563** is located at upper left-hand corner of the image **560**.

(102) The programming of FIG. **17** is executed independently for each lane **556**, **558**. The programming begins at a block **580** that instructs a first printing device comprising a portion of the system **20**, such as the second imager unit **44**, to print registration marks or fiducials **584** (one of which is shown in FIG. **15** and another is shown in FIG. **20**) on the first side **24a** of the web **24**, wherein each registration mark is printed together with one of the repeating printed content portions laid down by the unit **44** and is disposed at a controlled position **585** (one of which is seen in FIG. **18**) with respect to and adjacent such printed content portion. Specifically, as seen in the embodiment of FIGS. **19** and **20**, each registration mark **584** may be of any suitable design, such as, for example, three white dots arranged in a triangular configuration wherein a center of the three dots is disposed upstream and to the left at precise distances along the process direction and the

lateral direction, respectively, from the origin point **563** of what will become, when fully printed, an associated content portion, such as the image **560a** as shown in FIG. **20**. The registration marks **584** are, therefore, preferably printed outside of the web areas that are to be imaged.

(103) Referring again to FIG. **17**, the programming continues at a block **590** that senses the output of the camera **500** of FIG. **16** downstream of the imager unit **60** and upstream of the imager unit **70**. In the illustrated embodiment, the camera **500** comprises a CCD device or other suitable optical device that develops an optical reproduction of either the entire web **24**, an entire web portion **24a** and/or **24b**, or only a portion of each web portion **24a** or **24b**. Thus, in the illustrated embodiment, for example, the system **300** includes separate cameras **500** and **502**, although these cameras may be replaced by a single camera that simultaneously captures images of the laterally offset web sides **24a** and **24b**. In any event at least one camera is provided to sense each registration mark on each side of the web **24**. When the camera **500** detects a center point of a registration mark, a block **592** determines any physical offset of the center point in the X-direction and the Y-direction from an expected position. The pixel buffer(s) **307**, which may include one or more output lane ring buffers, are prestored with the raster-image processed (RIP) data for several content portions to be next imaged and intervening blank portions in the associated lane. In this regard it may be noted that the output lane ring buffer(s) continuously output data on a sequential raster-by-raster basis for the content portions and the intervening blank portions. If the block **592** determines that position corrections are necessary, a block **594** sequentially offsets pointers ("X, Y indexes") associated with the RIP data in a first raster for the next content portion to be imaged by the imager unit **70**. A block **596** monitors the offset process, and when the offset process for the last of the RIP data of the first raster has been completed, the pointers for the first raster are used by a block **598** to deliver the RIP data for the first raster at the required offset, which is determined by counting pulses developed by the position encoder **160**, to an output buffer of the pixel buffers **307**. The blocks **594**, **596**, and **598** continually operate to offset the pointers for subsequent rasters of RIP data and deliver such data to the output buffer. Next, a block **600** delays the delivery of the RIP data to the imager unit **70** by a time that takes into account the distance of the registration mark from the leading edge of the content portion to be next printed by the imager unit **70** and the speed of the web as detected by the position encoder **160** and a block **602** transmits the RIP data to the unit **70** at the proper time so that the content portion is printed accurately on the web **24**.

(104) Control from the block **602** returns to the block **590** to await the sensing of the next registration mark.

(105) As noted previously, the programming to reproduce content portions in the lane **558** is identical to that shown and described above and such programming is executed independently from the programming of FIG. **17**. In fact, as shown in FIG. **23**, more lanes, such as lanes **610**, **612**, **614**, **616**, and **618** may each be printed by an instance of the programming of FIG. **17** wherein the programming instances operate independently.

(106) FIG. **10** illustrates an embodiment in which registration is undertaken for both sides of the web **24a**, **24b**. Once the first side **24a** is imaged as noted above, the web is turned upside down as noted previously and traverses a second, laterally offset path during the second pass. In one embodiment the sensor **504** detects the registration mark **584** through the transparent web **24**. Alternatively, the sensor **504** may be disposed below the web **24** and directly detect the registration mark **584**. In either case, an instance of the programming of FIG. **17** operates the imager unit **44** to print white content portion in a registered position on the web side **24b** together with another registration mark **589** similar or identical to the registration mark **584** both in terms of the configuration and placement relative to the content portion printed by the imager unit **44** this time on the second side of the web **24**. The camera **502** thereafter detects the registration mark **589** to operate the imager unit **70** in register with the white printed content applied by the imager unit **44**.

(107) If desired each lateral portion of each of the remaining imager units **30**, **60**, and **82** may be operated by independent instances of the programming of FIG. **5** so that overall imager unit to

imager unit registration is achieved, whether simplex printing or duplex printing.

(108) The pull module **22**, the web guides **42**, **48**, **66**, and **81**, and the rollers described above provide a web transport that conveys the web **24** past the imager units **44**, **60**, **70**, and **82**. In some embodiments, each of imager units **44**, **60**, and **82** comprises a inkjet print unit **1184**, **1186**, and **1188**, respectively, and a print unit controller **1190**, **1192**, and **1194**, respectively. Each inkjet print unit **1184**, **1186**, and **1188** is adapted to selectively deposit a particular material substantially along the width of the web **24**. In particular, each inkjet print unit **1184**, **1186**, and **1188** includes a sufficient number inkjet printheads so that the ejection nozzles of such inkjet printheads substantially span a width of the web **24**. In some embodiments, if the inkjet print unit **1184**, **1186**, or **1188** includes a plurality of inkjet printheads (rather than just one web-wide inkjet printhead), such plurality of inkjet printheads are disposed abutting one another end-to-end in linear fashion to span the web **24**. In other embodiments, such plurality of inkjet printheads may be disposed in a carrier (not shown) in a two-dimensional array of inkjet printheads so that the ejection nozzles of the inkjet printheads (and of the inkjet print unit **1184**, **1186**, or **1188** comprising such inkjet printheads) span the width of the web **24**.

(109) Further, the imager unit **70** includes a plurality of inkjet printing units **228a-228h** disposed around a circumference of the drum **72**. Each inkjet printing unit **228a-228h** includes a sufficient number of inkjet printheads such that the ejection nozzles of the inkjet printheads substantially span the width of the web **24**. The inkjet printhead(s) that comprise(s) each inkjet printing unit **228a-228h** is/are adapted deposit a particular material along substantially along the width of the web **24**. For example, the inkjet printhead(s) that comprise(s) the inkjet print unit **228a** are disposed so that such inkjet printhead(s) may deposit a cyan ink substantially along the width of the web **24**. Similarly, the inkjet printhead(s) that comprise(s) the inkjet print unit **228b-228h** are disposed such inkjet printhead(s) may deposit magenta, yellow, black, orange, violet, green, and a spot color ink, respectively.

(110) Similar to the arrangement of the inkjet printheads of the inkjet print units **1184**, **1186**, and **1188**, the inkjet printheads that comprise each inkjet print unit **228a-228h** may be disposed abutting one another end-to-end in linear fashion or in a two dimensional array such that the ejection nozzles of the inkjet printheads of each inkjet print group **228** span the width of the web **24**.

(111) Each inkjet print unit **1184**, **1186**, **1188**, and **228a-228h** is associated with a print unit controller **1190**, **1192**, **1194**, and **1196a-1196h**, respectively. Each print unit controller **1190**, **1192**, and **1194** receives, from the print management control system **130**, layer data **150a**, **150b** and **150d** to be printed by the print unit **1184**, **1186**, and **1188**, respectively, associated therewith and position information of where such layer data **150a**, **150b**, and **150d** should be printed. Each print unit controller **1190**, **1192**, and **1194** controls the inkjet print units **1184**, **1186**, and **1188**, respectively, to cause the nozzles of such print unit to eject ink (or other material) onto the web **24** in accordance with such layer data **150a**, **150b**, and **150d** and position data.

(112) Further, the print management control system **130** provides layer data **150c**, representing all of the color bitmaps to be printed using process color inks to the print unit controllers **1196a-1196h** and position information of where on the web **24** to print such layer data **150c**. In some embodiments, the layer data **150c** is provided in its entirety to all of the print unit controllers **1196a-1196h**. In response, the print unit controller **1196** selects the color bitmap from the layer data **150c** that is associated with the color of ink to be printed by the inkjet print unit **228**, and generates signals to cause the inkjet printheads of such print unit **228** to deposit drops of such color of ink in accordance with the selected bitmap and position data. In other embodiments, the print management control system **130** provides the bitmap from the layer data **150c** that is associated with the color of ink that is printed by the inkjet print unit **228** to the print unit controller **1196** associated with such inkjet print unit.

(113) In some embodiments, to support printing at high speeds, the positions of all of the inkjet print units (and the inkjet printheads) that comprises the imager units **44**, **60**, **70**, and **82** are fixed

(i.e., stationary) during printing as the web **24** is transported thereby.

(114) As discussed above, the imager units **44**, **60**, **70**, **82**, and thus the inkjet print units **1184**, **1186**, and **1188**, respectively, thereof, may be operated to deposit drops of ink or other material having different volumes and at different resolutions.

(115) In one embodiment, the imager unit **44** deposits a white (or other) colorant onto the clear web **24** to create a backing (or silhouette) onto which subsequent colorants may be deposited. Because the white colorant includes particles such as titanium dioxide, a relatively large drop volume (e.g., between about 5 pL and 12 pL/drop) is required to accommodate such particles. Further, because the silhouette comprises an image that has a substantially identical intensity level throughout, the silhouette may be formed at a relatively low resolution, for example, 600 dpi. Such large drop size and low resolution may also allow the drops of the material to coalesce and form a consistent layer of colorant to form the silhouette.

(116) As discussed above, the imager unit **60** deposits a metallic ink on top of the colorant deposited by the imager unit **44**. Like the colorant deposited by the imager unit **44**, the metallic ink typically includes colorants and other materials having a relatively large particle size and the metallic ink is deposited to form a printed image that has little variability in intensity. Thus, the image using the metallic ink may be formed using drops of relatively large volume (e.g., from about 5 pL/drop to 12 pL/drop) and at a relatively low resolution (e.g., about 600 dpi).

(117) The imager unit **70** forms a high-resolution color image on the web. Therefore, the imager unit **70** forms an image using each print unit **228** with drops of ink having a relatively low volume (e.g., between about 3 pL/drop and about 6 pL/drop) and at a high resolution (e.g., 1200 or more dpi). Such low drop volume and high resolution form an image that has intensity variability throughout to reproduce the page **150** with fine detail therein.

(118) In some embodiments, the layer data **150a-150d** generated by the ripping and color correction process (block **154**) is screened bitmap data and the inkjet print units **1184**, **1186**, **228**, and **1188** are controlled by the print unit controllers **1190**, **1192**, **1196**, and **1194**, respectively, to place drops of material on the web **24** in accordance with such screened bitmap data. In other embodiments, the bitmap data generated by the ripping and color correction process (block **154**) is not screened and the print unit controllers **1190**, **1192**, **1196**, and **1194** screen the bitmap data provided by the ripping and color correction process (block **154**) and drive the print units **1190**, **1192**, **1196**, and **1194** to deposit drops of material on the web **24** in accordance with the screened data developed by the print unit controllers **1190**, **1192**, **1196**, and **1194**.

(119) In some embodiments, the data used to drive the low-resolution inkjet print units **1184** and **1186** is screened in accordance with a conventional halftone (e.g., amplitude modulated) screening pattern. Further, the data used to drive the high-resolution inkjet print units **228** and **1188** is screened in accordance with a frequency modulated screening pattern. It would be appreciated by one who has ordinary skill in the art that using the frequency modulated screening pattern allows reproduction of greater detail printed using such pattern. Other screening methods apparent to one who has ordinary skill in the art may be also be used.

(120) In some embodiments, the print unit controllers **1190**, **1192**, **1196**, and **1194** operated on one or more computer processors separate from computer processors used to implement the print management control system **130**. In other embodiments, one or more of the print unit controllers **1190**, **1192**, **1196**, and **1194** may operate as processes on the computer processors used to implement the print management control system **130**.

(121) As noted above, in some embodiments, the imager unit **82** is used to deposit a coating material onto the image printed by the imager unit **70**. To ensure that a thin layer of coating is deposited, such imager unit also prints at high resolution using a relatively small drop volume.

(122) It should be apparent to one who has ordinary skill in the art that using inkjet printheads that print at a relatively low resolution using large drop sizes when possible may be more cost effective than using inkjet printheads that print at high resolution with small drop sizes. Further, one having

ordinary skill in the art would appreciate that the amount of data that has to be transmitted between the print unit controllers **1190** and **1192** and the inkjet print unit **1184** and **1186**, respectively, that print at low resolution/large drop size may be substantially less than the amount of data that has to be transmitted from the inkjet print controller **228** and the inkjet print unit **1196** that prints at high-resolution/small drop size, and thus the costs of implementing the print unit controllers **1190** and **1192** may be less than the cost of implementing the inkjet print controller **228**.

(123) Referring once again to FIG. **3**, a camera **162** may be disposed following the image unit **82** that, when used, images the entire width of the web **24** (54 inches in the illustrated embodiment) and allows the print management control system **130** (or any of the other control systems of the system **20**) undertake color-to-color registration and color calibration, detect and correct for missing pixel(s) and stitching errors (gaps or alignment errors between portions of an image printed by adjacent heads), and undertake printhead normalization across the web.

(124) In some embodiments, the print management control system **130** undertakes a distortion correction process (block **1200**) prior to undertaking the ripping and color correction process (block **154**). As described in greater detail below, the distortion correction process (block **1200**) adjusts the dimensions of the page **150** (or elements thereof) to compensate for shrinking of the portion of the web **24** on which such page **150** is printed when the portion of the web **24** is used in a shrink wrap application.

(125) FIGS. **24A** and **24B** illustrates the dimensional compensation performed by the distortion correction process (block **1200**) undertaken of the print management control system **130**. In the example shown in FIG. **24B**, assume that after printing, the web **24** is to be used to produce a shrink-wrapped package **1202** (i.e., after the web **24** is shrunk) having a first image **1204** having dimensions (x, y) and a second image **1206** having dimensions (w, z) printed thereon. The print management control system **130** undertakes distortion correction (block **1200**) and determines that to compensate for the shrinking of the film, the first image **1204** should be printed having dimensions (x', y') and the second image should be printed having dimensions (w', z').

(126) The distortion correction process (block **1200**) also determines dot gain changes that may result in each of the images **1204** and **1206** as a result shrinking the portions of the web **24**, where such images are printed for example, because the distance between the dried drops of ink on the web decreases when such portions are shrunk. Thereafter, the distortion correction process (block **1200**) adjusts the image data to be printed to compensate for such dot gain changes prior to providing such image data to the ripping and color correction process (block **154**).

(127) Referring also to FIG. **25**, the distortion correction process (block **1200**) comprises a distortion corrector process (block **1232**), a page analyzer process (block **1234**), a distortion loader process (block **1236**), and a database **1238**. FIG. **26** shows a flowchart **1250** of the steps undertaken by the distortion correction process (block **1200**). Referring to FIGS. **3** and **24-26**, at step **1252**, the distortion corrector process (block **1232**) loads a page file to print and printing parameters including the inks (or other materials) to be deposited by the imager units **44**, **60**, **70**, and **82**, the material of the web **24** to be printed on, a final product that the web **24** will be formed into (by shrinking), and the like.

(128) At step **1254**, the distortion loader process (block **1236**) queries the database **1238** for distortion information data in accordance with the job parameters. In particular, such distortion information data includes information regarding dimensional changes different portions of the material of the web **24** undergoes when shrunk. For example, a portion of the web **24** proximate an outer edge of the web may shrink more (or less) compared to a portion of the web **24** proximate a central portion of the web. In some embodiments, such dimensional change information includes changes that occur when the web is shrunk to positions of a grid of equally spaced horizontal and vertical lines on an unshrunk web **24**. The equally spaced horizontal and vertical lines define a two-dimensional array of cells that comprise the grid. Each cell of the grid is associated with a portion of the web **24** on which an image may be printed and represents a predetermined area of contiguous

pixels of such image. For example, each cell of the grid may represent a portion of an image that is 32 pixels wide by 32 pixels high. It should be apparent to one who has ordinary skill in the art that each cell may represent portions of the image having other dimensions. Each cell of the grid is associated with distortion information that includes how a portion of an image that is to be printed on the portion of the web **24** associated with the cell is to be modified to compensate for distortion that may occur to such printed portion of the image due to shrinkage of the web after printing.

(129) The distortion information associated with each cell includes horizontal vertical scale factors by which the dimensions of the portion of the image to be printed on the portion of the web **24** associated with cell should be adjusted. In addition, the distortion information associated with each cell also includes information regarding adjustments that should be made to color values of pixels of the portion of the image associated with the cell to compensate for dot gain changes that may result from shrinking of the web **24** to dried drops of each type of ink (or other material) deposited on the web **24**.

(130) Further, the distortion information data may identify portions of the web **24** on which scannable elements (e.g., barcodes, QR codes, and the like) should not be printed because, for example, such portions may become too distorted or even occluded when the web **24** is shrunk around a product disposed therein. The distortion information may also identify alternate locations of the web **24** where such scannable elements should be repositioned if they happen to fall on a portion of the web **24** on which scannable elements should not be printed.

(131) At step **1256**, the page analyzer process (block **1234**) selects from the page file loaded at step **1252** a page element that is to be printed. Such page element may include an image, a scannable element, a text block, and the like. At step **1258**, the distortion corrector process (block **1232**) determines the position on the web **24** the selected page element is to be printed, uses the distortion information loaded at step **1254** and such position to determine dimensional changes to apply to the selected page element, and adjusts the dimensions (e.g., by resampling an image, adjusting font metrics, and the like) of the selected page element to develop an adjusted page element. The distortion corrector process (block **1232**) may also, at step **1258**, adjust the start position where adjusted page element is to be printed on the unshrunk web **24** in accordance with the distortion data.

(132) At step **1260**, the distortion corrector process (block **1232**) checks if the selected page element is a scannable element and the adjusted start position would place the printed scannable element on a portion of the web **24** where such scannable element should not be printed. If so, the distortion corrector process (block **1232**) proceeds to step **1262**, otherwise the distortion corrector proceeds to step **1264**.

(133) At step **1262**, the distortion corrector process (block **1232**) adjusts the position of scannable element (as adjusted at step **1258**) to an alternate location (e.g., as identified in the distortion data loaded as step **1254**) and proceeds to step **1264**.

(134) At step **1264**, the distortion corrector process (block **1232**) adjusts values of pixels of the adjusted page element to compensate for dot gain changes that may occur because of shrinking the web **24**. Alternately, for example, if the page element is not an image, the distortion corrector process (block **1232**) adjusts color values specified by print commands in the page file associated with the page element, as would be apparent to one who has ordinary skill in the art.

(135) At step **1266**, the distortion corrector process (block **1232**) adds the adjusted page element that results from dot gain compensation applied at step **1260** to an output page file and printing commands to cause the page element to be printed at a position on the web **24** determined at steps **1258** or **1262**.

(136) At step **1268**, the page analyzer process (block **1234**) determines if there any additional page elements that have not been processed and, if so, returns to step **1256** to select another page element. Otherwise, at step **1270**, the distortion loader process (block **1232**) adds the output page file to an input queue associated with the ripping and color correction process (block **154**, FIG. 3)

or otherwise provides the output page file to such process. Thereafter, the distortion correction block **1200** exits.

(137) Referring once again to FIGS. **3** and **25**, the distortion correction process (block **1200**) includes an on-press distortion analyzer process (block **1280**), an in-plant distortion analyzer process (block **1282**), and a customer site distortion analyzer process (block **1284**) that develop and adjust the distortion information stored in the database **1238**.

(138) FIG. **27** is a flowchart **1300** of steps undertaken by the on-press distortion analyzer process (block **1280**) to monitor distortion during a production run.

(139) Referring to FIG. **27**, the on-press distortion analyzer process (block **1280**), at step **1302**, loads parameters of a production job including the page **150** to be printed on the web **24**, the material that comprises the web **24**, and the like.

(140) At step **1304** the on-press distortion analyzer process (block **1280**) selects from the database **1238** the distortion information in accordance with the parameters of the production job.

(141) At step **1306**, the on-press distortion analyzer process (block **1280**) waits for the production job to begin.

(142) At step **1308**, the on-press distortion analyzer process (block **1280**) receives from a camera (not shown) disposed along a path of the web **24** between the dryer unit **84** and the take up roll **100** an image of a page printed on the web **24**. In some embodiments, the on-press distortion analyzer on-press distortion analyzer process (block **1280**) determines when a page will be in the field of view of the camera and directs the camera to acquire the image. In other embodiments, the camera acquires images of all pages printed on the web at a predetermined rate in accordance with the web speed and page size being printed. Other ways of operating the camera to acquire the image of the printed page apparent to one who has ordinary skill in the art may be used.

(143) At step **1310**, the on-press distortion analyzer on-press distortion analyzer process (block **1280**) analyzes the image of the printed page relative to the page data **150** (FIG. **3**) used to generate the printed page to estimate distortion that has occurred during printing.

(144) At step **1312**, the on-press distortion analyzer on-press distortion analyzer process (block **1280**) determines if the amount of distortion (either in dimensions of the printed page or in dot gain) determined at step **1310** exceeds a predetermined acceptable level of distortion, and, if so, the on-press distortion analyzer on-press distortion analyzer process (block **1280**), at step **1314**, generates an error to the print management control system **130** to stop the production run because of excessive distortion and exits.

(145) Otherwise, at step **1316**, the on-press distortion analyzer on-press distortion analyzer process (block **1280**) adjusts the distortion information in the database **1238** associated with the parameters of the production run in accordance with the distortion determined at step **1310**.

(146) At step **1318**, the on-press distortion analyzer on-press distortion analyzer process (block **1280**) determines if the production run has completed, and if so exits. Otherwise, the on-press distortion analyzer on-press distortion analyzer process (block **1280**) proceeds to step **1308** to receive another image.

(147) The in-plant distortion analyzer process (block **1282**) analyzes an image of a representative bag formed from a web **24** to develop distortion information used by the distortion corrector process (block **1232**) and the on-press distortion analyzer on-press distortion analyzer process (block **1280**). In some embodiments, a model of a product that is to be placed in bags formed from web **24** may be placed in the representative bag and the representative bag may be shrunk therearound. FIG. **28** illustrates a flowchart **1350** of the steps undertaken by the in-plant distortion analyzer process (block **1282**).

(148) At step **1352**, the in-plant distortion analyzer process (block **1282**) loads the job parameters used to print the web **24** that was used to form the bag.

(149) At step **1354**, the in-plant distortion analyzer process (block **1282**) initializes new distortion information that is associated with the job parameters.

(150) At step **1356**, the in-plant distortion analyzer process (block **1282**) receives an image of the representative bag after the bag has been formed and heat shrunk.

(151) At step **1358**, the in-plant distortion analyzer process (block **1282**) identifies in the received image a printed page element printed on the bag and selects a page element in the page **150** that corresponds to the printed page element, by for example, comparing the contents and position of the printed page element to the specification of the page element in the page **150**. In addition, the in-plant distortion analyzer process (block **1282**) undertakes image processing operations such as edge detection, thresholding, and the like to isolate in the received image the printed page element from other portions of the received image.

(152) At step **1360**, the in-plant distortion analyzer process (block **1282**) determines the dimensional and position distortion between the printed page element identified at step **1356** and the page element in the page **150** corresponding thereto.

(153) At step **1362**, the in-plant distortion analyzer process (block **1282**) updates the distortion information created at step **1354** with the dimensional and position distortion determined at step **1356** and associates such distortion with the position on the unshrunk web **24** where the page element was printed (as specified in the page **150**).

(154) At step **1364**, the in-plant distortion analyzer process (block **1282**) determines if all of the printed page elements in the image received at step **1356** have been analyzed, and, if so, proceeds to step **1366**. Otherwise, the in-plant distortion analyzer process (block **1282**) proceeds to step **1358** to identify another printed element.

(155) At step **1366**, the in-plant distortion analyzer process (block **1282**) stores the distortion information developed in steps **1362** and **1364** in the database **1238**, and then exits.

(156) Referring once again to FIG. **25**, the customer-site distortion analyzer process (block **1284**) is used to update distortion information stored in the database **1238** in accordance with information received after a product has been placed in the bag created from the web **24**, and the bag is shrunk around the product.

(157) FIG. **29** shows a flowchart **1400** of steps undertaken by a bag loading system to prepare data for use by the customer-site distortion analyzer process (block **1284**). At step **1402**, a product is placed in a bag produced from the printed web **24**.

(158) At step **1404**, the bag having the product therein is shrunk (e.g., in a heated water bath or other method apparent to one who has ordinary skill in the art).

(159) At step **1406**, a scannable print element on the shrunk bag is scanned.

(160) At step **1408**, data (e.g., a SKU or other identifying information) that results from scanning the scannable print element and an image of the scannable print element are stored on a computer (not shown). The computer may be at the production facility where the customer-site distortion analyzer process (block **1284**) operates, on a computer in the cloud, or at any other location accessible to the customer-site distortion analyzer process (block **1284**).

(161) At step **1410**, a bag loading system determines if additional bags remain to be loaded with product and scanned, and if so, proceeds to step **1402**. Otherwise, the bag loading system exits.

(162) Periodically, for example, after a predetermined number of production runs to produce bags in which a particular type of product is to be disposed, the customer-site distortion analyzer process (block **1284**) operates to determine if distortion errors are causing scanning errors at step **1408** (FIG. **29**).

(163) FIG. **30** is a flowchart **1450** of steps undertaken by the customer-site distortion analyzer process (block **1284**) to update distortion information to reduce scanning errors. At step **1452**, the customer-site distortion analyzer process (block **1284**) selects images of scannable page elements that are associated with mis-scans. Such scannable page elements may have encoded therein information regarding when and where the web **24** from the bag imprinted was printed with the scannable page element, a sequence code, and the other production information.

(164) At step **1454**, the customer-site distortion analyzer process (block **1284**) loads the job

parameters, the page **150**, and the distortion information associated with the job during which the printed scannable item that resulted in the mis-scan was printed on the web **24**.

(165) At step **1456**, the customer-site distortion analyzer process (block **1284**) analyzes each image selected at step **1452** with respect to the scannable page element in the page **150** to determine the distortion present in the selected image.

(166) At step **1458**, the customer-site distortion analyzer process (block **1284**) updates the distortion information loaded at step **1454** and associated with the job parameters in accordance with the distortion determined at step **1456**. At step **1460**, the customer-site distortion analyzer process (block **1284**) stores the updated distortion in the database **1238** for use with subsequent jobs having job parameters identical to those loaded at step **1454**.

(167) Thereafter, the customer-site distortion analyzer process (block **1284**) exits.

(168) FIGS. **31A** and **31B** graphically illustrates an example of how dimensional distortion information may be developed at step **1310** (FIG. **27**), step **1360** (FIG. **28**), and step **1456** (FIG. **30**).

(169) Referring to FIG. **31A**, a first two-dimensional array **1500** of cells **1502** is created wherein each cell spans a predetermined number of pixels of an element to be printed. For clarity, the reference number **1502** associated with each cell of grid is shown in FIG. **31A** with only a few such cells.

(170) Preferably, each cell **1502** of the first two-dimensional array **1500** spans an equal number of pixels horizontally and vertically. An image element **1504** in the page **150** to be printed is associated with the two-dimensional array of cells **1502** to determine the number of cells **1502** spanned by the image element **1504**. In the example shown in FIG. **31A**, the image **1504** spans an area of 5 cells horizontally and 5 cells vertically.

(171) Referring to FIG. **31B**, after the image element **1504** is printed on the web **24**, an image **1506** of the printed page element is acquired after the web **24** has been shrunk and formed into a bag. The acquired image is aligned with a second two-dimensional array **1510** of cells **1512**. Initially, the dimensions of each cell **1512** is identical to the dimension of each cell **1510**.

(172) Thereafter, the dimensions of the cells **1512** are adjusted so that the acquired image **1506** spans an identical number of cells **1502** spanned by the image **1504** (i.e., 5×5). The number of horizontal pixels spanned by the adjusted cell **1512** divided by the number of horizontal pixels spanned by the cell **1502** provides a horizontal scaling factor. Similarly, the number of vertical pixels spanned by the adjust cell **1512** divided by the number of vertical pixels spanned by the cell **1502** provides a vertical scaling factor. Such horizontal and vertical scaling factors are stored in the distortion information in the database **1238**.

(173) As discussed above, each cell **1502** is associated with a predetermined area of pixels of a portion of the image **1504**. The changes to such portion of the image associated with each cell may be analyzed as described above to determine the dimensions of each adjusted cell **1512**. Horizontal and vertical scale factors may be calculated from such determined dimensions to and stored as distortion information associated with each cell **1502**. Similarly, changes to image density (i.e., dot-gain) due to shrinking of the web **24** may be analyzed to determine the dot-gain adjustment needed to compensate for such changes and also stored as distortion information associated with each cell **1502**.

(174) Referring to FIG. **32**, each dryer unit **32**, **46**, **64**, **80**, and **84** associated with an imager unit **30**, **44**, **60**, **70**, and **82**, respectively, comprises a closed-loop dryer controller **2202**, an encoder roller **2204**, one or more heater unit(s) **2206a-2206n**, one or more temperature sensing devices **2208a-2208n**, a roller **2210**, and a camera **2212**.

(175) After the web **24** is printed on by the imager unit **30**, **44**, **60**, **70**, or **82**, as described above, the web **24** is conveyed past the encoder roller **2204** that generates a plurality of signals, one such signal for each revolution undertaken thereby. The imager unit **70** includes a plurality of printheads **228a-228h** that, for example, deposit process and/or spot color inks onto the web **24**.

(176) Each heater unit **2206a-2206n** is associated with a temperature sensing device **2208a-2208n**, respectively, and the heater unit(s) **2206a-2206n** and the temperature sensing device **2208a-2208n** are disposed such that the web **24** is conveyed therebetween. Further, each heater unit **2206** generates a flow of heated air that is blown toward a side **2214** of the web **24** having material deposited thereon by the imager unit **30, 44, 62, or 68**. In a preferred embodiment, the direction of the flow of heated air is perpendicular to the side **2214** of the web **24**. However, the flow of heated air may be directed toward the web at other angles or even transverse to the web to heat the web.

(177) The closed-loop dryer controller **2202** monitors the drying of the material on the web **24** and an indication of a temperature of the web **24** developed by the temperature sensing device **2208** to ensure that the material is sufficiently dried and that the temperature of the web **24** does not become so great as to damage the web (e.g., cause the web to shrink.) All of closed-loop dryer controllers **2202** of the system **20** are configured prior to a production run by a global dryer control system **2216** in accordance with parameters of the production run. The global dryer control system **2216** and the closed-loop dryer controller **2202** comprise the closed-loop dryer management system **2217** noted above.

(178) After the web **24** passes between the heater unit(s) **2206a-2206n** and the temperature sensing device(s) **2208**, the web is conveyed past the roller **2210** and the camera **2212**. The roller **2210** is the first roller (or any other component of the dryer unit **32, 46, 64, 80, 84**) that contacts the side **2214** of the web **24**, and thus any material deposited on such side **2214**. The roller **2210** may be an idler roller that supports the web **24**, a chiller roller that facilitates cooling of the web, or any other type of roller or component that first contacts the side **2214** after the web **24** has been conveyed past the heater unit(s) **2206a-2206n**. The camera **2212** is positioned to capture one or more image(s) of the side **2214** as the web **24** is conveyed thereby.

(179) At the beginning of a production run (or print job), the global dryer control system **2216** receives information regarding the production run from a data system **2218** and configures the closed-loop dryer controller **2202** with a minimum temperature the web **24** must reach to dry material deposited thereon by the imager unit **30, 44, 60, 70, or 82** associated with the dryer unit **32, 46, 64, 80, or 84**, respectively, and a maximum temperature that a temperature of the web cannot exceed to ensure the web does not undergo undesired shrinking or other damage. The global dryer control system **2216** also determines a maximum speed at which the web **24** may be conveyed to ensure that the web **24** has sufficient heater dwell time (i.e., exposure to the flow(s) of heated air generated by heater unit(s) **2206a-2206n**) to dry the deposited material and configures a transport control **2220** to set the conveyance speed of the web **24**.

(180) FIG. 32A illustrates a computer system **2230** especially adapted to implement the closed-loop dryer management system **2217**, it being understood that any or all of the control systems disclosed herein, such as one or more of the control systems **120, 130, 2218**, and/or **2220**, may be implemented by like computer systems or by the computer system **2230**. Thus, for example, the computer system **2230** may also comprise one or more processing unit(s) **2232** and may implement the closed-loop dryer management system **2217**. Each processing unit **2232** comprises a personal computer, server, or other programmable device having a memory **2234** that, among other things, stores programming executed by one or more processing module(s) or controller(s) **2236** to implement the closed-loop dryer management system **2217**. One or more of the processing unit(s) **2232** receive(s) signals from the temperature sensing device(s) **2208** and other sensors, receive(s) signals from the web position encoder **2204**, controls operation of heater units **2206** and/or a blower **2482** of the dryer units **32, 46, 64, 80, or 84** and the camera **2212**, and communicates with the supervisory control **120**, the data system **2218**, and transport control **2220**.

(181) FIG. 33 is a flowchart **2250**, of the steps undertaken by the global dryer control system **2216** to configure the closed-loop dryer controller **2202** and the transport control **2220**. Specifically, at step **2252**, the global dryer control system **2216** receives, from the data system **2218**, information regarding the production run including, for example, characteristics of the substrate that comprises

the web **24**, a desired web conveyance speed, characteristics of the material deposited by each imager unit **30**, **44**, **62**, and **68**, resolution and drop sizes each imager unit **30**, **44**, **62**, and **68** is to deposit, and the content to be printed.

(182) At step **2254**, the global dryer control system **2216** analyzes the content to be printed, the resolution to be printed by each imager unit **30**, **44**, **62**, and **68**, and the drop sizes that such imager units are configured to deposit to develop an estimate of a maximum material volume on any portion of the web **24** that is to be deposited by any of the imager units **30**, **44**, **62**, and **68**. Such maximum material volume may be represented as a dot-percent of material, a volume of material per area of the web, or another metric apparent to one who has ordinary skill in the art.

(183) In some embodiments, the maximum material volume per area of the web **24** is calculated by another system (not shown) when the content is prepared for printing and stored in the data system **2218**. In such embodiments, the global dryer control system **2216** receives the maximum material volume per area from the data system, at step **2254**.

(184) At step **2256**, the global dryer control system **2216** determines, based on the characteristics of the substrate that comprises the web **24**, a maximum temperature such substrate may reach without being damaged. In some embodiments, the data system **2218** includes such maximum temperature information for each type of substrate and the global dryer control system **2216** retrieves such information.

(185) In a preferred embodiment, the maximum web temperature determined at step **2256** is less than a temperature that would cause shrinkage or other harm to the web **24**. For example, if a particular substrate that comprises the web **24** begins to shrink at a temperature of 130° F. (about 54° C.), the maximum web temperature may be set to 125° F. (about 52° C.).

(186) Referring once again to FIG. **33**, at step **2258**, the global dryer control system **2216** determines the minimum web temperature the web **24** will have to reach in order to sufficiently dry the maximum material volume per area determined at step **2254**. In some embodiments, the data system **2218** includes information, for each type of material, the temperature a particular volume of such material must reach to be dried. In such embodiments, the global dryer control system **2216** uses such material information and maximum material volume per area to determine the minimum web temperature the web will have to reach.

(187) In a preferred embodiment, the minimum web temperature determined at step **2258** is greater than the temperature at which the maximum volume of material per area that is to be deposited during the production run would dry completely. For example, if the maximum volume of material per area to be deposited for the production run would dry completely at a temperature of 115° F., the minimum web temperature may be set to 120° F.

(188) In other embodiments, the global dryer control system **2216** may calculate the minimum web temperature in accordance with the maximum web temperature determined at step **2256** by, for example, multiplying the maximum web by a predetermined value greater than zero and less than 1. In some embodiments, such predetermined value between is about 0.90 to about 0.98. In other embodiments, such predetermined value is between about 0.85 and about 0.98, and still other embodiments, such predetermined value is between about 0.95 and about 0.97.

(189) In some embodiments, the global dryer control system **2216** calculates one minimum web temperature in accordance with the maximum volume of material per area that is deposited by all of the imager units **30**, **44**, **60**, **70**, and **82**. In other embodiments, the global dryer control system **2216** calculates a minimum web temperature for each dryer unit **32**, **46**, **64**, **80**, or **84** in accordance with a maximum volume of material per area that is expected to be deposited by the imager unit **30**, **44**, **60**, **70**, or **82**, respectively, associated with such heater unit **2206**.

(190) At step **2260**, the global dryer control system **2216** calculates a necessary web speed that will provide sufficient heater dwell time for the web to reach the minimum web temperature estimated at step **2258**. In some embodiments, the data system **2218** provides information regarding the dwell time and temperature necessary for the material on the web **24** to sufficiently dry and the data

system **2218** or global dryer control system **2216** determines the necessary web speed to provide such dwell time based on the material comprising the web **24** and the heating characteristics of the heater units **2206**.

(191) At step **2262**, the global dryer control system **2216** determines if the web speed calculated at step **2260** is less than or equal to the desired web conveyance speed loaded at step **2252**. If so, the global dryer control system **2216** configures the transport control **2220** to set the web speed for the production run to the desired web conveyance speed, at step **2264**. Otherwise, at step **2266**, the global dryer control system **2216** configures the transport control **2220** to set the web speed for the production run to the necessary web speed calculated at step **2260**.

(192) At step **2270**, the global dryer control system **2216** configures the closed-loop dryer controller **2202** of each dryer unit **32**, **46**, **64**, **80**, and **84** in accordance the minimum and maximum web temperatures determined at steps **2258** and **2256**, respectively.

(193) At step **2272**, the global dryer control system **2216** determines a location along the width of the web **24** that is to receive the maximum material volume per area calculated in step **2254**. In some embodiments, the global dryer control system **2216** operates a camera **2212** positioning apparatus (not shown) to automatically position the camera **2212** so that the camera **2212** is able to capture such determined location. In other embodiments, the global dryer control system **2216** informs an operator to manually position the camera **2212** so the camera **2212** can capture the determined location. Thereafter, the global dryer control system **2216** exits.

(194) Referring once again to FIG. **32**, after the closed-loop dryer controller **2202** and the transport control **2220** have been configured by the global dryer control system **2216**, and the production run started, each closed-loop dryer controller **2202** operates the heating unit(s) **2206** associated therewith to maintain the temperature of the web **24** between the minimum and maximum temperatures. In addition, the closed-loop dryer controller **2202** detects if the material deposited on the web **24** is not being dried sufficiently and, for example, pick off is occurring and, in response, adjusts the heating unit(s) **2206** associated therewith and/or the transport control **2220** accordingly.

(195) FIG. **34** shows a flowchart **2300** of the steps undertaken by the closed-loop dryer controller **2202** to maintain the temperature of the web **24** and to detect and prevent insufficient drying.

Referring to FIG. **34**, at step **2302**, the closed-loop dryer controller **2202** loads the minimum and maximum temperature information determined by the global dryer control system **2216** at steps **2256** and **2258** (FIG. **33**).

(196) At step **2304**, the closed-loop dryer controller **2202** selects which ones of the heating unit(s) **2206a-2206n** available in the drying unit **32**, **46**, **64**, **80**, or **84** will be operated to maintain the temperature of the web **24** at least at the minimum temperature during the production run. In some embodiments, the dryer unit **32**, **46**, **64**, **80**, **84** may be configured with only one heater unit **2206**. In other embodiments, the dryer unit **32**, **46**, **64**, **80**, **84** may be configured with as many as 18 (or more) heater units **2206** and only a subset of such heater units may be used during the production run. In situations, where heavy material coverage is expected or a slow drying material is deposited on the web **24**, all of the available heater units **2206** may be used. In some embodiments, all of the heater units **2206** may be used when the production run is started and the number of heater units **2206** may be adjusted during the production run in response to monitoring of the temperature of the web **24**.

(197) At step **2308**, the closed-loop dryer controller **2202** determines a temperature and a speed of the flow of heated air generated by each selected heater unit **2206** during the production run. For example, a first one of the selected heater units (e.g., heater unit **2206a**) that the web **24** passes after having been printed on may be configured to direct the flow of heated air toward the side **2214** of the web **24** at a lower speed and higher temperature than a subsequent heater unit **2206**. It should be apparent to one of ordinary skill the art that the material deposited on the web **24** is relatively fluid when the web **24** reaches the first heater unit **2206a** and that directing the flow of heated air at a high speed may disturb such material. As the material dries, the flow of heated air may be

directed at the web at higher speeds without disturbing the material.

(198) In some embodiments, the closed-loop dryer controller **2202** sets the speed of the heated air generated by the first heater unit **2206a** to be between about 0.1 and about 0.2 cubic feet per minute per linear inch of the width of the web **24**. Such air flow speed may be incrementally increased at one or more subsequent heater units **2206b** through **2206n** until the speed of the heated air generated by the heater unit **2206** that is operated and is most distal from the imager unit **30**, **44**, **60**, **70**, or **82** is approximately 2 cubic per minute per linear inch of a width the web **24**.

(199) Further, it should be apparent to one who has ordinary skill in the art, that evaporation of solvent in the material as the web **24** passes past the heater units **2206** facilitates cooling of the web **24**. Thus, the flow of heated air generated by the first heater unit **2206a** toward the web **24** may have a higher temperature because the solvent content of the material exposed to such flow of heated air is highest relative to when the material is exposed to air from subsequent heater unit(s) **2206b-2206n**.

(200) In some embodiments, the flow of heated air generated by the first heater unit **2206a** exceeds the temperature at which the web **24** begins to shrink (i.e., a shrink temperature). For example, if the shrink temperature of the web is 130° F. (about 54° C.), the temperature of the flow of heated air generated by the first heater unit **2206a** may be set to about 190° F. (about 88° C.). Further, the temperature of the airflow generated by subsequent heater unit(s) **2206b-2206n** may ramp downward so that the airflow generated by the heater unit **2206** most distal from the imager unit **30**, **44**, **60**, **70**, or **82** is near the shrink temperature of the web (or less).

(201) At step **2310**, the closed-loop dryer controller **2202** configures each heater unit **2206** selected at step **2306** to generate the flow of heated air in accordance with the speed and temperature determined at step **2308** for that heater unit **2206**.

(202) At step **2312**, the closed-loop dryer controller **2202** waits to receive a job start signal, for example, from the supervisory control system **120** (FIG. 1), that indicates that the production run is to begin. Also at step **2312**, the closed-loop dryer controller **2202** directs the heater unit(s) **2206** selected at step **2306** to begin generating the flow of heated air.

(203) At step **2314**, the closed-loop dryer controller **2202** polls the temperature sensing devices **2208** associated with the heater units **2206** being used for the production run to acquire a temperature of the web **24** sensed by each temperature sensing device **2208**.

(204) At step **2316**, the closed-loop dryer controller **2202** determines whether insufficient drying of the material may be occurring, as described in greater detail below.

(205) At step **2318**, the closed-loop dryer controller **2202** determines if the web temperature sensed by any of the temperature sensing devices polled at step **315** exceeds the maximum web temperature loaded at step **2302** and, if so, proceeds to step **2320**. Otherwise, the closed-loop dryer controller **2202** proceeds to step **2322**.

(206) At step **2320**, the closed-loop dryer controller **2202** adjusts operation of the heater unit(s) **2206** to facilitate reducing the temperature of the web **24** and then proceeds to step **2324**.

(207) At step **2322**, the closed-loop dryer controller **2202** checks if the temperature of the web **24** determined at step **2314** is too low for the material deposited thereon to dry or if insufficient drying of the material was determined at step **2316** and, if so, the closed-loop dryer controller **2202** proceeds to step **2324**. Otherwise, the closed-loop dryer controller **2202** proceeds to step **2326**. In particular, the closed-loop dryer controller **2202** analyzes the temperatures of the web **24** sensed by all of the temperature sensing device **2208** and if none of the sensed temperatures of the web **24** exceed the minimum web temperature, the closed-loop dryer controller **2202** determines that web temperature is too low.

(208) At step **2324**, the closed-loop dryer controller **2202** adjusts operation of the heater unit(s) **2206** to facilitate raising the temperature of the web **24**, and then proceeds to step **2326**.

(209) At step **2326**, the closed-loop dryer controller **2202** determines if a job send signal has been received from the supervisory control system **120**. If such signal has not been received, the closed-

loop dryer controller **2202** returns to step **2314**. Otherwise, the closed-loop dryer controller **2202** initiates a shutdown process for the heater units **2206** and exits.

(210) FIG. **35** shows a flowchart **2350** of steps undertaken at step **2316** (FIG. **34**) by the closed-loop dryer controller **2202** to determine if the material on the web **24** is insufficiently dried. Referring to FIGS. **32** and **35**, as described above, insufficient drying of the web may be detected when the material deposited on the side **2214** of the web **24** contacts a roller, e.g. roller **2210**, before such material is fully dried. A portion of the undried material is transferred to the roller **2210**, and then from the roller **2210** to a subsequent portion of the side **2214** of the web **24**.

(211) Referring to FIG. **35**, at step **2352**, the closed-loop dryer controller **2202** determines if the camera **2212** has acquired an image of the web **24** is available for analysis. If no such image has been acquired, the closed-loop dryer controller **2202** proceeds to step **2354**, otherwise the closed-loop dryer controller **2202** proceeds to step **2356**.

(212) At step **2354**, the closed-loop dryer controller **2202** analyzes the content that is to be printed to determine a first time in the future when an image will be printed by the imager unit **30**, **44**, **62**, **68** on a first portion of the web **24** and that will in the field of view of the camera **2212**. At step **2358**, the closed-loop dryer controller **2202** uses the frequency of the signals generated by the encoder roller **2204** (FIG. **32**) and a predefined circumference of the encoder roller **2204** to determine the speed of the web **24**.

(213) At step **2360**, the closed-loop dryer controller **2202** determines in accordance with the first time and the web speed, a second time when a second portion of the web **24** immediately following the first portion of the web **24** and that is supposed to be free of material will be in the field of view of the camera **2212**.

(214) At step **2362**, the closed-loop dryer controller **2202** set a trigger to cause the camera **2212** to acquire an image of the second portion of the web **24** at the second time and store such image in a memory location accessible by closed-loop dryer controller **2202** and the camera **2212**. In one embodiment, at step **2362**, the closed-loop dryer controller **2202** sets a timer that causes an interrupt to be generated at the second time. In addition, the closed-loop dryer controller **2202** associates an image capture process to be initiated when such interrupt is generated. Such image capture process directs the camera **2212** to acquire the image, receives the acquired image, and stores the acquired image in the shared memory. Other ways of triggering the camera **2212** to capture an image at particular time apparent to one who has ordinary skill in the art may be used.

(215) After the trigger has been set at step **2362**, the closed-loop dryer controller **2202** proceeds to step **2316** of FIG. **34**.

(216) If, at step **2352**, the closed-loop dryer controller **2202** determines that an image is available for analysis (i.e., an image acquired in response to the trigger set at step **2362** being actuated), the closed-loop dryer controller **2202**, at step **2356**, analyzes the acquired image. As described above, the captured image is of the second portion of the web **24** that is expected to be free of any material. The closed-loop dryer controller **2202** analyzes the captured image to determine if any pixels thereof have a value that indicates that material has been transferred to the second portion of the web **24**. For example, the closed-loop dryer controller **2202** may apply a threshold operation to the acquired image that selects pixels having intensity values greater than a predetermined intensity value. If at least a predetermined number of pixels are selected as a result of such threshold operation, then the closed-loop dryer controller **2202** determines that material transfer from the roller **2210** to the second portion has occurred. Otherwise, the closed-loop dryer controller **2202** determines that no such material transfer has occurred. It should be apparent that other ways of analyzing the captured image to determine whether material transfer has occurred apparent to one who has ordinary skill in the art may be used. After undertaking step **2356**, the closed-loop dryer controller **2202** proceeds to step **2316** of FIG. **34**.

(217) FIG. **36** is a flowchart **2400** of the steps undertaken by the closed-loop dryer controller **2202** to reduce the temperature of the web **24**. Referring to FIG. **36**, the closed-loop dryer controller

2202, determines, at step **2402**, if the speed of the flow of heated air can be adjusted to reduce the temperature of the web **24**. If so, then at step **2404**, the closed-loop dryer controller **2202** directs the one or more of the heater units **2206a-2206n** to reduce the speed of the flow of heated air of generated thereby, and thus reduce the convection of heat from the such heater units **2206** to the web **24**. After undertaking step **2404**, the closed-loop dryer controller **2202** proceeds to step **2324** (FIG. **34**).

(218) If at step **2402**, the closed-loop dryer controller **2202** determines that the speed of the flow of heated air cannot be adjusted, then, at step **2406**, the closed-loop dryer controller **2202** determines if the temperature of the heated air generated by one or more heater unit(s) **2206a-2206n** can be reduced. For example, if the all of the heater unit(s) **2206a-2206n** are operating at their minimum operating temperature, then such temperature cannot be reduced.

(219) If the temperature of the flow of heated air can be reduced, then at step **408**, the closed-loop dryer controller **2202** selects a heater unit **2206** and directs such heater unit **2206** to generate the flow of heated air at a lower temperature. In one embodiment, the closed-loop dryer controller **2202** selects the heater unit **2206** operating at the highest temperature and reduces the temperature of such heater unit **2206** by a predetermined amount (e.g., 5° F.) or by a percentage of the current setting of the temperature of the flow of heated air (e.g., 10%). In other embodiments, the closed-loop dryer controller **2202** selects and reduces the temperature of the flow of heated air generated by the heater unit **2206** most distal to the imager unit **30, 44, 62, or 68**. After undertaking step **2408**, the closed-loop dryer controller **2202** proceeds to step **2324** (FIG. **34**). It should be that other ways to select the heater unit **2206** to adjust in this manner and/or amount of such adjustment apparent to one who has ordinary skill in the art may be used.

(220) If at step **2406**, the closed-loop dryer controller **2202** determines that the temperature of one of the heater unit(s) **2206a-2206n** cannot be reduced, the closed-loop dryer controller **2202** determines, at step **2410**, if more than one heater unit **2206a-2206n** is operating and, if so, whether one such heater unit **2206** can be turned off. If so, then at step **2412**, the closed-loop dryer controller **2202** turns off the heater unit **2206** most distal, most proximal, or intermediate the most distal and most proximal from the imager unit **30, 44, 62, or 68**. After undertaking step **2412**, the closed-loop dryer controller **2202** proceeds to step **2324** (FIG. **34**). In an exemplary embodiment the closed-loop dryer controller **2202** turns off the heater unit **2206** that is operating and is most distal from the image unit **30, 44, 62, or 68**.

(221) If at step **2410**, the closed-loop dryer controller **2202** determines that one of the heater unit(s) **2206a-2206n** cannot be turned off, the closed-loop dryer controller **2202**, at step **2414** determines if the conveyance speed of the web **24** can be increased (e.g., if the web **24** is not being conveyed at maximum speed) to reduce the heater dwell time of the web **24**. If so, the closed-loop dryer controller **2202** directs the transport control **2220** to increase the web speed, at step **2416**. After undertaking step **2416**, the closed-loop dryer controller **2202** proceeds to step **2324** (FIG. **34**).

(222) If at step **2414**, the closed-loop dryer controller **2202** determines that the web speed cannot be increased, then, in some embodiments, the closed-loop dryer controller **2202**, at step **2418**, generates an error signal to, for example, the supervisory control system **120** that the temperature of the web **24** cannot be reduced and an operator should be alerted and/or a shutdown procedure started. Thereafter, the closed-loop dryer controller **2202** proceeds to step **2324** (FIG. **34**).

(223) FIG. **37** is a flowchart **2450** of the steps undertaken by the closed-loop dryer controller **2202** to raise the temperature of the web **24**. Referring to FIG. **37**, the closed-loop dryer controller **2202**, at step **2452**, determines if the speed of the flow of heated air can be adjusted to raise the temperature of the web **24**. If so, then at step **2454**, the closed-loop dryer controller **2202** increases the speed of the flow of heated air of one or more of the heater unit(s) **2206a-2206n** to increase the convection of heat from the such heater unit(s) **2206**. After undertaking step **2454**, the closed-loop dryer controller **2202** proceeds to step **2324** (FIG. **34**).

(224) Otherwise, at step **2456**, the closed-loop dryer controller **2202** determines if the temperature

of the flow of heated air generated by one or more heater units **2206a-2206n** can be increased. For example, if the all of the heater unit(s) **2206a-2206n** are operating at their maximum operating temperature, then such temperature cannot be increased.

(225) If the temperature of the flow of heated air can be increased, then at step **2458**, the closed-loop dryer controller **2202** selects a heater unit **2206** and directs such heater unit **2206** to generate the flow of heated air at a higher temperature. In one embodiment, the closed-loop dryer controller **2202** selects the heater unit **2206** operating at the lowest temperature and increases the temperature of such heater unit **2206** by a predetermined amount (e.g., 5° F.) or by a percentage of the current setting of the temperature of the flow of heated air (e.g., 10%). In other embodiments, the closed-loop dryer controller **2202** selects and increases the temperature of the flow of heated air generated by the heater unit **2206** most proximal to the imager unit **30**, **44**, **62**, or **68**. After undertaking step **2454**, the closed-loop dryer controller **2202** proceeds to step **2324** (FIG. **34**). Other ways to select a heater unit **2206** to adjust in this manner and/or amount of such adjustment apparent to one who has ordinary skill in the art may be used.

(226) If at step **2456**, the closed-loop dryer controller **2202** determines that the temperature of the flow of air generated by any of the heater unit(s) **2206a-2206n** cannot be raised to increase the temperature of the web **24**, the closed-loop dryer controller **2202** determines, at step **2460** if all of the heater units **2206a-2206n** are operating or if an additional heater unit **2206** can be turned on. If an additional heater unit **2206** can be turned on, then at step **2462**, the closed-loop dryer controller **2202** turns on an additional heater unit **2206**. After undertaking step **2462**, the closed-loop dryer controller **2202** proceeds to step **2324** (FIG. **34**). In some embodiments, the closed-loop dryer controller **2202** turns on the heater unit **2206** that is not operating and that is most distal, most proximate, or intermediate from the imager unit **30**, **44**, **62**, or **68**. In an exemplary embodiment, the closed-loop dryer controller **2202** turns on the heater unit **2206** that is not operating and that is most proximate the imager unit **30**, **44**, **62**, or **68**.

(227) If at step **2460**, if the closed-loop dryer controller **2202** determines that all of the heater units **2206a-2206n** are operating, the closed-loop dryer controller **2202**, at step **2464** determines if the conveyance speed of the web **24** can be decreased to increase the heater dwell time of the web **24**. If so, the closed-loop dryer controller **2202** directs the transport control **2220** to reduce the web speed, at step **2466**. After undertaking step **2466**, the closed-loop dryer controller **2202** proceeds to step **2324** (FIG. **34**).

(228) If, at step **2464**, the closed-loop dryer controller **2202** determines that the web speed cannot be reduced, then, in some embodiments, the closed-loop dryer controller **2202**, at step **2468**, generates an error signal to, for example, the supervisory control system **120** that the temperature of the web **24** cannot be increased and an operator should be alerted and/or a shutdown procedure started. Thereafter, the closed-loop dryer controller **2202** proceeds to step **2324** (FIG. **34**).

(229) Referring once again to FIG. **32**, in some embodiments each heater unit **2206** is coupled by a corresponding air duct to a turbo-blower unit **2482**. The turbo-blower unit **2482** supplies a flow of unheated air to all of the heater unit(s) **2206a-2206n**, which in turn heat such flow of unheated air and to create the flow of heated air directed toward the web **24**. In some embodiments, the closed-loop dryer controller **2202** adjusts the speed of the flow of unheated air generated by turbo-blower unit **2482** to increase or decrease the speed of the flow of heated air generated by all of the heater unit(s) **2206a-2206n**. In addition, the closed-loop dryer controller **2202** may individually adjust a heater unit **2206** to increase or decrease the speed of the flow of heated air generated thereby independently of the other heater units **2206**.

(230) In some embodiments, the temperature sensing device **2208** may be a temperature sensor that directly senses the temperature of the web **24** to develop an indication of the temperature of the web **24**. However, in some cases it may not always be feasible to directly sense the temperature of the web **24**. For example, a contact temperature sensor may interfere with conveyance of the web **24**. However, a contactless temperature sensor, e.g., an infrared temperature sensor, may not

accurately sense the temperature of the web **24** because, for example, the web **24** has portions that are clear or has material disposed thereon that is of varying colors and/or comprises one or more metallic component(s). FIGS. **38A** and **38B** illustrate two embodiments of temperature sensing devices **2208** that use a contact less temperature sensor **2480** to develop an indication of the temperature of the web **24**.

(231) Referring to FIG. **38A**, the temperature sensing device **2208** includes a heat-conductive roller **2483**, such as an idler roller, disposed opposite the heater unit **2206** and the web rides on such heat-conductive roller **2483**. The heat-conductive roller **2483** is heated by the web **24** and the temperature sensor **2480** monitors the temperature of the heat-conductive roller **2483** to develop an indication of the temperature of the web **24**.

(232) Alternately, referring to FIG. **38B**, instead of the roller **2483**, the temperature sensing device **2208** includes a heat-conductive plate **2484** disposed opposite the heater unit **2206** and the web **24** is conveyed past such plate **2484**. The heat-conductive plate **2484** is heated by the web **24** and the temperature sensor **2480** monitors the temperature of the heat-conductive plate **2484** to develop an indication of the temperature of the web **24**. It should be apparent that in such embodiments, the temperature sensor **2480** may be a contact less sensor or may be a contact sensor attached to the plate **2484**.

(233) Other configurations and ways of operating the temperature sensing device **2208** to develop an indication of the temperature of the web **24** apparent to those who have ordinary skill in the art may be used.

(234) In some embodiments, additional sensors may be disposed in or proximate the dryer unit **32**, **46**, **64**, **80**, or **84** to sense the ambient conditions proximate thereto. For example, a humidity sensor (not shown) may be disposed proximate the dryer unit **32**, **46**, **64**, **80**, or **84** to sense the humidity proximate thereto and the global dryer control system **2216** and/or the closed-loop dryer controller **2202** may use information from such additional sensors to adjust the speed and/or temperature of the airflow generated by the heater unit(s) **2206**.

(235) Referring to FIG. **32**, the dryer unit **32**, **46**, **64**, **80**, or **84** may include additional components including for example one or more roller(s) (e.g., roller **490**) or other components (not shown) to guide and/or support the web **24** as it is conveyed through such dryer unit.

(236) In some embodiments, the global dryer control system **2216** may receive information from the closed-loop dryer controller **2202** regarding whether the initial necessary web speed and minimum temperature developed at the start of a particular production run did not result in material deposited on the web **24** being sufficiently dried. The global dryer control system **2216** may adjust the information in the data system **2218** that a slower web speed and/or higher temperature should be used for other production runs that have characteristics similar to the particular production run.

(237) In some embodiments, the global dryer control system **2216** may monitor the content that is going to be printed by the imager unit **30**, **44**, **60**, **70**, or **82** during a production run. If the global dryer control system **2216** determines that the characteristics of such content will result in a substantially more or less volume of the material being deposited on the web **24**, the global dryer control system **2216** may develop an updated necessary web speed and/or minimum temperature the web **24** should reach and reconfigure the closed-loop dryer system in accordance with such updated web speed and temperature.

(238) It should be apparent to those who have skill in the art that any combination of hardware and/or software may be used to implement any or all of the system or components thereof described herein. It will be understood and appreciated that one or more of the processes, sub-processes, and process steps described in connection with the FIGS. may be performed by hardware, software, or a combination of hardware and software on one or more electronic or digitally-controlled devices. The software may reside in a software memory (not shown) in a suitable electronic processing component or system such as, for example, one or more of the functional systems, controllers, devices, components, modules, or sub-modules schematically

depicted in the FIGS. The software memory, for example the memory **304**, may include an ordered listing of executable instructions for implementing logical functions (that is, “logic” that may be implemented in digital form such as digital circuitry or source code, or in analog form such as analog source such as an analog electrical, sound, or video signal). The instructions may be executed within the processing module or controller **306** that includes, for example, one or more microprocessors, general purpose processors, combinations of processors, digital signal processors (DSPs), field programmable gate arrays (FPGAs), or application-specific integrated circuits (ASICs). Further, the block diagrams describe a logical division of functions having physical (hardware and/or software) implementations that are not limited by architecture or the physical layout of the functions. The example systems described in this application may be implemented in a variety of configurations and operate as hardware/software components in a single hardware/software unit, or in separate hardware/software units.

(239) The executable instructions may be implemented as a computer program product having instructions stored therein which, when executed by a processing module of an electronic system, direct the electronic system to carry out the instructions. The computer program product may be selectively embodied in any non-transitory computer-readable storage medium for use by or in connection with an instruction execution system, apparatus, or device, such as an electronic computer-based system, processor-containing system, or other system that may selectively fetch the instructions from the instruction execution system, apparatus, or device and execute the instructions. In the context of this document, computer-readable storage medium is any non-transitory means that may store the program for use by or in connection with the instruction execution system, apparatus, or device. The non-transitory computer-readable storage medium may selectively be, for example, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device. A non-exhaustive list of more specific examples of non-transitory computer readable media include: an electrical connection having one or more wires (electronic); a portable computer diskette (magnetic); a random access, i.e., volatile, memory (electronic); a read-only memory (electronic); an erasable programmable read only memory such as, for example, flash memory (electronic); a compact disc memory such as, for example, CD-ROM, CD-R, CD-RW (optical); and digital versatile disc memory, i.e., DVD (optical).

(240) It will also be understood that receiving and transmitting of signals or data as used in this document means that two or more systems, devices, components, modules, or sub-modules are capable of communicating with each other via signals that travel over some type of signal path. The signals may be communication, power, data, or energy signals, which may communicate information, power, or energy from a first system, device, component, module, or sub-module to a second system, device, component, module, or sub-module along a signal path between the first and second system, device, component, module, or sub-module. The signal paths may include physical, electrical, magnetic, electromagnetic, electrochemical, optical, wired, or wireless connections. The signal paths may also include additional systems, devices, components, modules, or sub-modules between the first and second system, device, component, module, or sub-module.

(241) As should be evident from the foregoing, a printing composition such as a primer formulation may be applied directly to the surface of the web **24** or may be applied atop a coated surface, such as a white ink and/or metallic ink printed on the web **24**, so as to simultaneously enable the adherence of a printed image to the web **24**, and provide a chemical interaction with the overprinted inks optimizing color fidelity and overall image quality. Additionally, a printing composition such as an ink formulation may be applied directly to the surface of the web **24** or may be applied atop a coated surface, such as a primer and/or metallic ink, printed on the web **24** so as to simultaneously enable the adherence of a printed image to the web **24**, and provide a chemical interaction with the printed compositions optimizing color fidelity and overall image quality. An optional printing composition such as an overprint varnish may be applied to modify the surface finish and appearance of the final imaged web **24**.

(242) The printing composition may be a water-based primer applied by a flexographic or flood coating fashion to the web **24** by the first imager unit **30**. The printing composition comprises a pigment opacifier, for example, a dispersion of white colorant such as titanium dioxide, and therefore provides both a backing for the subsequently printed color ink jet image as well as opacity for preventing light pass-through on the reverse side of the web **24**. The printing composition also includes one or more polymers or one or more polymer dispersion, a surfactant, a defoamer agent or a defoamer agent dispersion, a surface treatment agent, and water. In an exemplary embodiment, a carrier of the pigment opacifier dispersion, polymer dispersion, and/or defoamer agent dispersion comprises water, preferably deionized water.

(243) Printing Composition 1

(244) In an exemplary embodiment, the printing composition includes from about 10.00% to about 30.00% by weight of a polymer or an equal weight in a polymer dispersion, more preferably from about 15.00% to about 25.00% by weight, and most preferably about 20.00% by weight of a polymer or an equal weight in a polymer dispersion of the total printing composition percentage. The printing composition also includes from about 0.25% to about 2.00% by weight of a surfactant, more preferably from about 0.50% to about 1.50% by weight, and most preferably about 1.00% by weight of a surfactant of the total printing composition percentage. The printing composition further includes from about 0.05% to about 0.25% by weight of a defoamer agent or an equal weight in a defoamer agent dispersion, more preferably from about 0.10% to about 0.20% by weight, and most preferably about 0.15% by weight of a defoamer agent or an equal weight in a defoamer agent dispersion. The printing composition also includes up to about 5.00% by weight of a surface treatment agent, more preferably up to about 3.00%, and most preferably up to about 1.00% of a surface treatment agent by weight of the total printing composition percentage. The printing composition further includes up to about 35.00% by weight a pigment opacifier or an equal weight in a pigment opacifier dispersion of the total printing composition percentage. The printing composition also includes water, such as deionized water, in addition to any water apart of any dispersion, in an amount by weight to sum to 100.00%.

(245) In an exemplary embodiment, the polymer or polymer dispersion, particularly for flexographic use is cationic, preferably a cationic acrylic colloidal dispersion that enables adherence of the printing composition and ink jet ink(s) to the substrate while simultaneously providing a chemical interaction with the variably printed ink jet ink optimizing print dot formation, size, color fidelity, and overall image quality.

(246) In an exemplary embodiment, an alkali resistant cationic acrylic copolymer emulsion, such as Ottopol K-23, available from Gellner Industrial of Tamaqua, Pennsylvania, may be mixed with water to create a cationic acrylic colloidal dispersion. The polymer dispersion acts as a binder to stabilize pigment in variably printed ink while providing long-term durability to the final product.

(247) In alternative embodiments, the polymer or polymer dispersion may be Ottopol K-362 or Ottopol K-633, or may be nonionic, such as Ottopol K-502, Takelac™ WS-635, available from Mitsui Chemicals of Tokyo, Japan, or Takelac™ WS-6355. In alternative embodiments, the polymer or polymer dispersion may be any other cationic or nonionic polymer or polymer dispersion, such as a food packaging compliant cationic or nonionic polymer or polymer dispersion.

(248) In an exemplary embodiment, the surfactant comprises a web wetting agent, preferably Surfynol® 465, available from Evonik Industries of Essen, Germany.

(249) In alternative embodiments, the surfactant may instead be another ethoxylated acetylenediol based surfactant such as Surfynol® 420, 440, or 485. Other surfactants, such as ethoxylated alcohol-based surfactants may instead be used. Alternate embodiments may instead comprise Surfynol® 104 PG 50, or any other cationic or nonionic surfactant, such as a food packaging compliant cationic or nonionic surfactant.

(250) In an exemplary embodiment, the defoamer agent or defoamer agent dispersion comprises an

emulsion of polysiloxanes, hydrophobic solids, and emulsifiers such as BYK®-023, available from BYK-Chemie GmbH of Wesel, Germany.

(251) In alternative embodiments, the defoamer agent or defoamer agent dispersion may instead be any other cationic or nonionic defoamer agent or defoamer agent dispersion, such as food packaging compliant cationic or nonionic defoamer agent or defoamer agent dispersion.

(252) In an exemplary embodiment, the surface treatment agent may comprise one or more soluble salts, such as multivalent salts or salts wherein the cation is selected from Mg^{2+} , Ca^{2+} , Ba^{2+} , Zn^{2+} , and Al^{3+} , more preferably Ca^{2+} or Mg^{2+} , and most preferably Ca^{2+} in combination with suitable counter ions. In this embodiment, calcium chloride is the surface treatment agent, since calcium chloride enhances the chemical neutralization and placement of the pigment solids contained in ink in contact with the printing composition.

(253) In an alternative embodiment, the surface treatment agent may comprise of any other soluble salts, such as a food packaging compliant multivalent salt.

(254) In an exemplary embodiment, the pigment opacifier or pigment opacifier dispersion, particularly for flexographic use comprises Ti-Pure™ R-900, available from the Chemours Company of Wilmington, Delaware, and Disperbyk®-190, available from BYK-Chemie GmbH.

(255) In an alternative embodiment, the pigment opacifier or pigment opacifier dispersion may instead comprise any other pigment opacifier, such as a food packaging compliant pigment opacifier.

(256) In an embodiment, the printing composition may further comprise a high surface area solid, such as silica or alumina colloidal particles. In this embodiment the amount of water by weight is reduced so as to have the addition of the high surface area solid with the other components to sum to 100.00%.

(257) The overall viscosity of the printing composition deposited via flexographic printing technologies will vary depending upon the configuration of the system **20**. In an exemplary embodiment, the viscosity is from about 40 centipoise (“cP”) to about 200 cP, more preferably from about 55 cP to about 150 cP, and most preferably from about 70 cP to about 120 cP.

(258) In another embodiment, the printing composition may be deposited using ink jet printer head(s) to the web **24** via the first imager unit **30**. Thereinafter, the second imager unit **44** may deposit metallic ink(s) onto at least portions of the web **24** that received the primer via the first imager unit **30**. A third imager unit **60** may deposit one or more primary process and/or secondary process color ink(s) on at least portions of the web that received the printing composition via the first imager unit **30**.

(259) Printing Composition 2

(260) In another embodiment, the printing composition may be deposited using ink jet printer head(s) to the web **24** using the third imager unit **60**. In this embodiment, a white pigmented ink is deposited via the first imager unit **30**, and one or more metallic ink(s) is/are deposited by the second imager unit **44** on at least portions of the web **24** that received the white pigmented ink from the first imager unit **30**. Then the water-based printing composition is deposited using the third imager unit **60** on at least portions of the web **24** that received the white pigmented ink from the first imager unit **30**.

(261) A water-based printing composition applied via ink jet as noted above is transparent and comprises a viscosity modifier, viscosity modifier dispersion, or viscosity modifier solution, a polymer, polymer dispersion, or polymer solution, a surfactant, a defoamer agent, defoamer agent dispersion, or defoamer agent solution, an antimicrobial agent, antimicrobial agent dispersion, or an antimicrobial agent solution, a surface treatment agent, and water. In an exemplary embodiment, a carrier of the viscosity modifier dispersion or viscosity modifier solution, polymer dispersion or polymer solution, defoamer agent dispersion, and/or antimicrobial agent dispersion or antimicrobial agent solution comprises water, preferably deionized water. In an exemplary embodiment, the printing composition includes from about 1.75% to about 3.25% by weight of a viscosity modifier

or an equal weight in a viscosity modifier dispersion or viscosity modifier solution, preferably from about 2.00% to about 3.00%, and most preferably from about 2.25% to about 2.75% by weight of a viscosity modifier or an equal weight in a viscosity modifier dispersion or viscosity modifier solution of the total printing composition percentage. The printing composition includes from about 1.00% to about 4.00% by weight of a polymer or an equal weight in a polymer dispersion or polymer solution, more preferably from about 1.50% to about 30.50%, and most preferably from about 20.00% to about 30.00% by weight of a polymer or an equal weight in a polymer dispersion or polymer solution of the total printing composition percentage. The printing composition includes from about 0.25% to about 2.00% by weight of a surfactant, more preferably from about 0.50% to about 1.75%, and most preferably from about 0.75% to about 1.50% by weight of a surfactant of the total printing composition percentage. The printing composition includes up to about 0.01% by weight of a defoamer agent or an equal weight in a defoamer agent dispersion of the total printing composition percentage. The printing composition includes up to about 0.02% by weight of an antimicrobial agent or an equal weight in an antimicrobial agent dispersion or antimicrobial agent solution of the total printing composition percentage. The printing composition includes up to about 70.50% by weight of a surface treatment agent, more preferably up to about 6.50% by weight, and most preferably up to about 5.00% by weight of a surface treatment agent of the total printing component percentage. The printing composition also includes water, such as deionized water, in addition to any water apart of any dispersion, in an amount by weight to sum to 100.00%.

(262) In an exemplary embodiment printing composition, particularly for use with ink jet printing, the viscosity modifier comprises a polyether polyol solution. A polyether polyol solution may be non-ionic, such as Rheolate® 350 D, available from Elementis of London, United Kingdom, and may be mixed in equal parts with water to create a VOC-free polyether polyol based associative rheology modifier for water-based compositions.

(263) In an alternative embodiment, the viscosity modifier may instead comprise any other cationic or nonionic viscosity modifier, such as a food packaging compliant cationic or nonionic polyether polyol solution.

(264) Also in an exemplary embodiment, the polymer or polymer dispersion or polymer solution comprises a cationic acrylic resin solution. An alkali resistant cationic acrylic resin solution, preferably Ottopol K-633, may be mixed with water to create a cationic low molecular weight acrylic resin solution.

(265) In alternative embodiments, the polymer or polymer dispersion or polymer solution may be Ottopol K-362, Ottopol K-23 or may be nonionic, such as Ottopol K-502, or any other cationic or nonionic polymer or polymer dispersion or polymer solution, such as a food packaging compliant cationic or nonionic polymer or polymer dispersion or polymer solution.

(266) In an exemplary embodiment, the surfactant comprises a web wetting agent, preferably Surfynol® 465.

(267) In alternative embodiments, the surfactant may be another ethoxylated acetylenediol based surfactant such as Surfynol® 420, 440, or 485. Other surfactants, such as ethoxylated alcohol based surfactants may instead be used. Alternate embodiments may instead comprise Surfynol® 104 PG 50, or any cationic or nonionic surfactant, such as a food packaging compliant surfactant.

(268) In an exemplary embodiment, the defoamer agent may be a nonionic defoamer, preferably Surfynol® DF 110L.

(269) In alternative embodiments, the defoamer agent, defoamer agent dispersion, or defoamer agent solution may instead be any other defoamer agent or defoamer agent dispersion that comprises at least one of acetylenic diol and ethylene glycol. In alternative embodiments, the defoamer agent or defoamer agent dispersion may instead be any other cationic or nonionic defoamer agent or defoamer agent dispersion, such as a food packaging compliant cationic or nonionic defoamer agent or dispersion of a defoamer agent.

(270) In an exemplary embodiment, the antimicrobial agent may comprise of a biocide, preferably

Proxel GXL, available from Arch Chemicals, Inc. of Norwalk, Connecticut, wherein the preservation of water-based products is achieved against microbial organisms.

(271) In alternative embodiments, the antimicrobial agent, antimicrobial agent dispersion or antimicrobial solution may instead be any other antimicrobial agent, antimicrobial agent dispersion, or antimicrobial solution, such as a food packaging compliant antimicrobial agent, antimicrobial agent dispersion, or antimicrobial solution.

(272) In an exemplary embodiment, the surface treatment agent may comprise one or more of soluble salts, such as multivalent salts or salts wherein the cation is selected from Mg^{2+} , Ca^{2+} , Ba^{2+} , Zn^{2+} , and Al^{3+} , preferably Ca^{2+} or Mg^{2+} , and most preferably Ca^{2+} in combination with suitable counter ions. In this embodiment, calcium chloride is the surface treatment agent, since calcium chloride enhances the chemical neutralization and placement of the pigment solids contained in ink deposited on the printing composition.

(273) In an alternative embodiment, the surface treatment agent may comprise any other soluble salts, such as a food packaging compliant multivalent salt.

(274) In an alternative embodiment, the printing composition may further comprise a high surface area solid, such as silica or alumina colloidal particles. In this embodiment the amount of water by weight is reduced so as to have the addition of the high surface area solid with the other components to sum to 100.00%.

(275) The overall viscosity of the primer composition deposited via ink jet printing technologies will vary depending upon the print heads of the system **20**. For instance the print head(s) that deposit(s) the color ink(s), such as primary process color inks, may have a viscosity range from about 3 cP to about 5 cP. The print head(s) that deposit(s) the white pigment ink may have a viscosity range from about 5 cP to about 6 cP. In an exemplary embodiment the print heads used in the system **20** are piezoelectric, and have an exemplary viscosity from about 2 cP to about 10 cP, more preferably from about 3 cP to about 10 cP, and most preferably from about 5 cP to about 6 cP. A color index of white (white color index) is the lightest color and is achromatic (i.e. having no hue). Furthermore, white color index has a hue angle of about 0 degrees, a saturation of about 0%, and a brightness of about 100%. A color index of non-white (non-white color index) is any color that is not the lightest color and is not achromatic (i.e. having a hue). Additionally, non-white color index has a hue angle greater than about 0 degrees, a saturation greater than about 0%, and a brightness of less than about 100%.

(276) Whether deposited via flexographic, ink jet or other, the printing compositions disclosed above are suitable for use in the printing system **20** because it is a single composition containing both adhesive bonding and ink receptive properties thereby consolidating multi-part coatings, such as a tie/adhesive layer and an ink jet primer layer into a single treatment that can be applied to the web **24** using multiple different printing technologies such as jettable (i.e. ink jet) or conventional (i.e. flexographic anilox flood coating, flexographic spot coating), and other known printing technologies in the field of printing.

(277) In general, the printing composition being either cationic and/or nonionic are “opposite” in chemical charge/pH to the anionic ink jet inks, and generally will interact with the pigments of the ink jet inks to fix them on the web **24**.

(278) Furthermore, such printing compositions are suitable for use on a web **24** subject to dimensional modification, such as heat-shrinkable film, due to the exemplary ability to dry and/or cure at low drying temperatures up to the threshold for dimensional integrity of the web **24** while the web **24** maintains integrity and dimensions. For example, the threshold for dimensional integrity of the web **24** may be 120° F. Additionally, the optimal drying and/or curing at such low drying temperatures allow the primer formulations to be used for variable ink jet printing with a high throughput. The throughput varies from run to run or within a single run in the range from about 0 fpm to about 1000 fpm, most preferably about 500 fpm. The ability to use such printing compositions at varying throughput makes short-run printing operation and market-segment

targeting more economically feasible.

(279) Furthermore, once the printing compositions are applied, an optional printing composition, such as an overprint varnish, may be applied to modify the surface finish and appearance of the final imaged web.

(280) Printing Composition 3

(281) The printing composition may be a water-based white ink applied by ink jet to the web **24** by the first imager unit **30**. The printing composition comprises a pigment or pigment dispersion, a polymer or polymer dispersion, a co-solvent, a surfactant, and water. In an exemplary embodiment, a carrier of the pigment dispersion and/or the polymer dispersion comprises water, preferably deionized water. In an exemplary embodiment, the co-solvent is water miscible and is a solvent or a carrier. In an exemplary embodiment, the printing composition includes from about 10.00% to about 14.00% by weight of a pigment or an equal weight in a pigment dispersion, preferably from about 10.50% to about 13.50% by weight, and most preferably from about 11.00% to about 13.00% by weight of a pigment or an equal weight in a pigment dispersion of the total printing composition percentage. The printing composition further includes from about 3.00% to about 7.00% by weight of a polymer or an equal weight in a polymer dispersion, preferably from about 3.50% to about 6.50% by weight, and most preferably from about 4.00% to about 6.00% by weight of a polymer or an equal weight in a polymer dispersion of the total printing composition percentage. The printing composition also includes from about 15.00% to about 19.00% by weight of a co-solvent, preferably from about 15.50% to about 18.50% by weight, and most preferably from about 16.00% to about 18.00% by weight of a co-solvent of the total printing composition percentage. The printing composition includes from about 0.20% to about 0.40% by weight of a surfactant, preferably from about 0.25% to about 0.35% by weight, and most preferably about 0.30% by weight of a surfactant of the total printing composition percentage. The printing composition also includes water, such as deionized water, in addition to any water apart of any dispersion, in an amount by weight to sum to 100.00%.

(282) In an exemplary embodiment, the pigment or pigment dispersion comprises a titanium dioxide pigment, preferably Jetsperse® AQ White (AQD-PWHT), available from Sun Chemical of Parsippany, New Jersey.

(283) In alternative embodiments, the pigment or pigment dispersion may be any other anionic pigment or pigment dispersion, such as a food packaging compliant anionic pigment or pigment dispersion.

(284) In an exemplary embodiment, the polymer or polymer dispersion is anionic and comprises a waterborne polyurethane dispersion (“PUD”), preferably Takelac™ WPB-341, available from Mitsui Chemicals of Tokyo, Japan.

(285) In alternative embodiments, the polymer or polymer dispersion may be any other nonionic or anionic polymer or polymer dispersion, such as a food packaging compliant nonionic or anionic polymer or polymer dispersion.

(286) In an exemplary embodiment, the co-solvent comprises an alcohol-based co-solvent, preferably 3-methoxy-3-methyl-1-butanol, available from Kuraray of Okayama, Japan.

(287) In alternative embodiments, the co-solvent may be any other co-solvent, such as a food packaging compliant co-solvent.

(288) In an exemplary embodiment, the surfactant comprises a web wetting agent that is solvent-free. The surfactant may be a polyether modified polysiloxane compound, preferably BYK® 345, available from BYK-Chemie GmbH of Wesel, Germany.

(289) In alternative embodiments, the surfactant may instead be another BYK® surfactants, such as BYK® 347, BYK® 348, BYK® 349 or may be another surfactant such as Dynol™ 980, available from Evonik of Essen, Germany, or instead be any other surfactant, such as a food packaging compliant surfactant.

(290) Printing Composition 4

(291) The printing composition may be a water-based non-white ink applied by ink jet to the web **24** by the third imager unit **60**. In this exemplary printing composition includes from about 3.00% to about 6.00% by weight of a pigment or an equal weight in a pigment dispersion, preferably from about 3.50% to about 5.50% by weight, and most preferably from about 4.00% to about 5.00% by weight of a pigment or an equal weight in a pigment dispersion of the total printing composition percentage. The printing composition includes from about 7.00% to about 11.00% by weight a polymer or an equal weight in a polymer dispersion or polymer solution, preferably from about 7.50% to about 10.50% by weight, and most preferably from about 8.00% to about 10.00% by weight of a polymer or an equal weight in a polymer dispersion or polymer solution of the total printing composition percentage. The printing composition further includes from about 13.00% to about 17.00% by weight of a co-solvent, preferably from about 13.50% to about 16.50% by weight, most preferably from about 14.00% to about 16.00% by weight of a co-solvent of the total printing composition percentage. The printing composition also includes from about 0.25% to about 2.00% by weight of a surfactant, preferably from about 0.50% to about 1.50% by weight, and most preferably about 1.00% by weight of a surfactant of the total printing composition percentage. The printing composition also includes water, such as deionized water, in addition to any water apart of any dispersion, in an amount by weight to sum to 100.00%. In an exemplary embodiment, a carrier of the pigment dispersion, and/or the polymer dispersion or the polymer solution comprises water, preferably deionized water. In an exemplary embodiment, the co-solvent is water miscible and is a solvent or a carrier.

(292) In an exemplary embodiment, the pigment or pigment dispersion comprises Pro-Jet™ pigment APD 1000, available from FujiFilm Imaging Colorants Inc. of New Castle, Delaware. The pigment dispersions may be primary process color inks such as APD 1000 Cyan dispersion, APD 1000 Magenta dispersion, APD 1000 LF Yellow dispersion, and APD 1000 Black.

(293) In alternative embodiments, other pigment dispersions may instead be APD 1500 Black, APD 5000 Black, or any other primary process or secondary process color, or spot color pigment compositions, such as food packaging compliant primary process color or secondary process color, such as orange, violet, and green, or spot color compositions.

(294) In an exemplary embodiment, the polymer, polymer solution, or polymer dispersion comprises a rosin adduct ester-based component, preferably a modified rosin solution derived from Lawter™ Filtrez™ 526A, available from Lawter Inc. of Chicago, Illinois. In the present application, a rosin derived from Lawter™ Filtrez™ 526A may be mixed with ammonium hydroxide and water to create a modified rosin solution that may be applied in composition via ink jet.

(295) In alternative embodiments, the polymer, polymer solution, or polymer dispersion may be any other food packaging compliant nonionic or anionic polymer, polymer solution, or polymer dispersion.

(296) In an exemplary embodiment, the co-solvent may be alcohol-based, preferably 3-methoxy-3-methyl-1-butanol.

(297) In alternative embodiments, the co-solvent may be any other co-solvent, such as a food packaging compliant co-solvent.

(298) In an exemplary embodiment, the surfactant comprises a web wetting agent that is solvent-free. The surfactant may be a polyether modified polysiloxane compound, preferably BYK® 345.

(299) In alternative embodiments, the surfactant may instead be another BYK® surfactants, such as BYK® 347, BYK® 348, BYK® 349 or may be another surfactant such as Dynol™ 980, or instead be any other food packaging compliant surfactant.

(300) The overall viscosity of the printing compositions disclosed above that may be deposited via ink jet printing technologies will vary depending upon the print head(s) of the system **20**. For instance the print head(s) that deposit(s) the printing composition, such as primary process color compositions, may have a viscosity range from about 3 centipoise (“cP”) to about 5 cP. The print

head(s) that deposit(s) the printing composition with a color index of white may have a viscosity range from about 5 cP to about 6 cP. In an exemplary embodiment the print heads used in the system **20** are piezoelectric and have an exemplary viscosity from about 2 cP to about 10 cP, more preferably from about 3 cP to about 10 cP, and most preferably from about 5 cP to about 6 cP. A color index of white (white color index) is the lightest color and is achromatic (i.e. having no hue). Furthermore, white color index has a hue angle of about 0 degrees, a saturation of about 0%, and a brightness of about 100%. A color index of non-white (non-white color index) is any color that is not the lightest color and is not achromatic (i.e. having a hue). Additionally, non-white color index has a hue angle greater than about 0 degrees, a saturation greater than about 0%, and a brightness of less than about 100%.

(301) Furthermore, the printing compositions disclosed above are suitable for use on a web **24** subject to dimensional modification, such as heat-shrinkable film, due to the exemplary ability to dry and/or cure at low drying temperatures up to the threshold for dimensional integrity of the web **24**, while the web **24** maintains integrity and dimensions. For example, the threshold for dimensional integrity of the web **24** may be 120° F. Within the threshold for dimensional integrity of the web **24**, the co-solvent in the above printing compositions is believed to have an evaporation profile that in combination with the other printing components allows for efficient drying and adherence of the printing components to the web **24**. Additionally within the threshold for dimensional integrity of the web **24**, the amount of polymer, polymer solution, or polymer dispersion in the above printing compositions is believed to be stable in a larger amount when added with the co-solvent and other components in the printing compositions. The high loading of the polymer, polymer solution, or polymer dispersion in the above printing compositions is believed to allow greater adherence of the printing composition to the web **24** or coated surface without diluting or decreasing the viscosity outside the viscosity range of the print head(s) of the system **20**. Additionally, the optimal drying and/or curing at such low drying temperatures allow the printing compositions to be used for variable ink jet printing with a high throughput. The throughput varies from run to run or within a single run in the range from about 0 fpm to about 1000 fpm, most preferably about 500 fpm. The ability to use such printing compositions at varying throughput makes short-run printing operation and market-segment targeting more economically feasible.

(302) Furthermore, once the printing compositions are applied, an optional printing composition, such as an overprint varnish, may be applied to modify the surface finish and appearance of the final imaged web.

(303) Printing Composition 5

(304) The printing composition may be a water-based overprint varnish applied by a flexographic or flood coating fashion to the web **24** by the fifth imager unit **82**. In an exemplary embodiment of an optional printing composition, particularly for use with flexographic technology, comprises a viscosity modifier or viscosity modifier dispersion, a polymer or polymer dispersion, a surfactant, a defoamer agent or defoamer agent dispersion, a surface additive, and water. In an exemplary embodiment, a carrier of the viscosity modifier dispersion, and/or the polymer dispersion comprises water, preferably deionized water.

(305) In an embodiment, the printing composition includes from about 1.00% to about 2.00% by weight of a viscosity modifier or an equal weight in a viscosity modifier dispersion, preferably from about 1.25% to about 1.75% by weight, most preferably about 1.50% by weight of a viscosity modifier or an equal weight in a viscosity modifier dispersion of the total printing composition percentage. The printing composition includes from about 8.00% to about 12.00% by weight of a polymer or an equal weight in a polymer dispersion, preferably from about 8.5% to about 11.50% by weight, most preferably from about 9.00% to about 11.00% by weight of a polymer or an equal weight in a polymer dispersion of the total printing composition percentage. The printing composition also includes from about 0.25% to about 2.00% by weight of a surfactant, more

preferably from about 0.50% to about 1.50% by weight, and most preferably about 1.00% by weight of a surfactant of the total printing composition percentage. The printing composition further includes from about 0.15% to about 0.45% by weight of a defoamer agent or an equal weight in a defoamer agent dispersion, more preferably from about 0.20% to about 0.40% by weight, and most preferably from about 0.25% to about 0.35% by weight of a defoamer agent or an equal weight in a defoamer agent dispersion of the total printing composition percentage. The printing composition includes from about 0.35% to 0.65% by weight of a surface additive, more preferably from about 0.40% to about 0.60% by weight, and most preferably from about 0.45% to about 0.55% by weight of a surface additive of the total printing composition percentage. The printing composition also includes water, such as deionized water, in addition to any water apart of any dispersion, in an amount by weight to sum to 100.00%.

(306) In an exemplary embodiment, the viscosity modifier or viscosity modifier dispersion comprises methylcellulose, preferably Methocel™ A15LV, available from DuPont de Nemours, Inc. of Wilmington, Delaware.

(307) In an alternative embodiment, the viscosity modifier or viscosity modifier dispersion may be any other viscosity modifier, such as a food packaging compliant viscosity modifier.

(308) In an exemplary embodiment, the polymer or polymer dispersion comprises a waterborne dispersion based on vinyl acetate and ethylene, such as Vinnapas® 410, available from Wacker Chemie AG of Munich, Germany.

(309) In an alternative embodiment, the polymer or polymer dispersion may be any other polymer or polymer dispersion, such as a food packaging compliant waterborne dispersion.

(310) In an exemplary embodiment, the surfactant comprises a web wetting agent, preferably Surfynol® 465.

(311) In alternative embodiments, the surfactant may be another ethoxylated acetylenediol based surfactant, or any other surfactant, such as a food packaging compliant surfactant.

(312) In an exemplary embodiment, the defoamer agent comprises a nonionic defoamer, preferably Surfynol® DF 110L.

(313) In alternative embodiments, the defoamer agent or defoamer agent dispersion may be any other cationic or nonionic defoamer agent or defoamer agent dispersion, such as a food packaging compliant cationic or nonionic defoamer agent or defoamer agent dispersion.

(314) In an exemplary embodiment, the surface additive comprises a silicone containing surface additive, such as BYK®-333.

(315) In alternative embodiments, the surface additive may be any other surface additive, such as a food packaging compliant surface additive.

(316) Printing Composition 6

(317) The printing composition may be a water-based overprint varnish applied by flexographic or flood coating fashion to the web **24** by the fifth imager unit **82**. In an embodiment of an optional printing composition, the printing composition comprises a viscosity modifier or viscosity modifier dispersion, a polymer or polymer dispersion, a surfactant, a defoamer agent or defoamer agent dispersion, a wax additive or wax dispersion, a polymer modifier or polymer modifier dispersion, a polyether siloxane copolymer, and water. In an exemplary embodiment, a carrier of the viscosity modifier dispersion, polymer dispersion, defoamer agent dispersion, wax additive dispersion, and/or polymer modifier dispersion comprises water, preferably deionized water.

(318) In an embodiment, the printing composition includes from about 1.00% to about 2.00% by weight of a viscosity modifier or an equal weight in a viscosity modifier dispersion, preferably from about 1.25% to about 1.75% by weight, most preferably about 1.50% by weight of a viscosity modifier or an equal weight in a viscosity modifier dispersion of the total printing composition percentage. The printing composition includes from about 9.00% to about 13.00% by weight of a polymer or an equal weight in a polymer dispersion, preferably from about 9.5% to about 12.50% by weight, most preferably from about 10.00% to about 12.00% by weight of a polymer or an equal

weight in a polymer dispersion of the total printing composition percentage. The printing composition also includes from about 0.25% to about 20.00% by weight a surfactant, more preferably from about 0.50% to about 1.50% by weight, and most preferably about 1.00% by weight of a surfactant of the total printing composition percentage. The printing composition includes from about 0.05% to about 0.25% by weight of a defoamer agent or an equal weight in a defoamer agent dispersion, more preferably from about 0.10% to about 0.20% by weight, and most preferably about 0.15% by weight of a defoamer agent or an equal weight in a defoamer agent dispersion of the total printing composition percentage. The printing composition includes from about 1.00% to 2.50% by weight of a wax additive, more preferably from about 1.25% to about 20.25% by weight, and most preferably from about 1.50% to about 20.00% by weight of a wax additive of the total printing composition percentage. The printing composition includes from about 0.25% to 1.75% by weight of a polymer modifier, more preferably from about 0.50% to about 1.50% by weight, and most preferably from about 0.75% to about 1.25% by weight of a polymer modifier of the total printing composition percentage. The printing composition includes from about 0.25% to 1.75% by weight of a polyether siloxane copolymer, more preferably from about 0.50% to about 1.50% by weight, and most preferably from about 0.75% to about 1.25% by weight of a polyether siloxane copolymer of the total printing composition percentage. The printing composition also includes water, such as deionized water, in addition to any water apart of any dispersion, in an amount by weight to sum to 100.00%.

(319) In an exemplary embodiment, the viscosity modifier or viscosity modifier dispersion comprises methylcellulose, preferably Methocel™ A15LV.

(320) In an alternative embodiment, the viscosity modifier or viscosity modifier dispersion may be any other viscosity modifier or viscosity modifier dispersion, such as a food packaging compliant viscosity modifier.

(321) In an exemplary embodiment, the polymer or polymer dispersion may be a waterborne dispersion based on vinyl acetate and ethylene, preferably Vinnapas® 410.

(322) In an alternative embodiment, the polymer or polymer dispersion may be any other polymer or polymer dispersion, such as a food packaging compliant waterborne dispersion.

(323) In an exemplary embodiment, the surfactant comprises a web wetting agent, preferably Surfynol® 465.

(324) In alternative embodiments, the surfactant may be another ethoxylated acetylenediol based surfactant such as Surfynol® 420, 440, and 485. Other surfactants, such as ethoxylated alcohol based surfactants may instead be used. Alternate embodiments may instead comprise Surfynol® 104 PG 50, or any other cationic or nonionic surfactant, such as a food packaging compliant cationic or nonionic surfactant.

(325) In an exemplary embodiment, the defoamer agent or defoamer agent dispersion comprises an emulsion of polysiloxanes, hydrophobic solids, and emulsifiers such as BYK®-023.

(326) In alternative embodiments, the defoamer agent or defoamer agent dispersion may be any other cationic or nonionic defoamer agent or defoamer agent dispersion such as a food packaging compliant cationic or nonionic defoamer agent or a defoamer agent dispersion.

(327) In an exemplary embodiment, the wax additive or wax additive dispersion comprises a wax additive of aqueous form, preferably Ultralube® E-912, available from Keim-Additec, of Kirchberg, Germany.

(328) In alternative embodiments, the wax additive or wax additive dispersion may be any other wax additive or wax additive dispersion, such as a food packaging compliant wax additive or wax additive dispersion.

(329) In an exemplary embodiment, the polymer modifier or polymer modifier dispersion comprises an associative rheological modifier, preferably Rheolate® 212.

(330) In alternative embodiments, the polymer modifier or polymer modifier dispersion may be any other rheological modifier, such as a food packaging compliant associative rheological modifier.

(331) In an exemplary embodiment, the polyether siloxane copolymer comprises TEGO® Glide 410, available from Evonik Industries.

(332) In an alternative embodiment, the polyether siloxane copolymer may be any other polyether siloxane copolymer, such as a food packaging compliant polyether siloxane copolymer.

(333) The following examples further illustrate the disclosure but should not be construed as in any way limiting its scope.

(334) Example 1. A printing composition useful in the present disclosure for flexographic application comprises the following formulation by weight: a. 35.00% by weight of an opacifier dispersion (TiO₂ dispersion including Ti-Pure™ R-900, Disperbyk®-190) (75% solids) b. 50.00% by weight of a polymer dispersion (Ottopol K-23) (42% solids) c. 1.00% by weight of a surfactant (Surfynol® 465) d. 0.80% by weight of a defoamer agent dispersion (BYK®-023) (19% solids) e. 1.00% by weight of a surface treatment agent (CaCl₂) f. 12.20% by weight of deionized water (DI H₂O)

(335) The 12.20% DI H₂O is additional deionized water to the water already used for the dispersions, such as Ottopol K-23 dispersion. The total amount of deionized water throughout the printing composition is 50.67% (counting dispersion and other subcomponent water content).

(336) Example 2. A printing composition according to the present disclosure for ink jet application comprises the following formulation by weight: a. 5.00% by weight of a viscosity modifier dispersion (Rheolate® 350 D) (50% solids) b. 10.00% by weight of a polymer dispersion (Ottopol K-633) (27% solids) c. 1.00% by weight of a surfactant (Surfynol® 465) d. 0.06% by weight of a defoamer agent dispersion (Surfynol® DF 110L) (20% solids) e. 0.10% by weight of an antimicrobial agent dispersion (Proxel GXL) (20% solids) f. 5.00% by weight of a surface treatment agent (CaCl₂) g. 78.84% by weight of deionized water (DI H₂O)

(337) The 78.84% DI H₂O is additional deionized water to the water already used for the dispersions, such as Ottopol K-633 dispersion. The total amount of deionized water throughout the primer formulation is 88.84% (counting dispersion and other subcomponent water content).

(338) Example 3. A printing composition useful in the present disclosure for ink jet application comprises the following formulation by weight: a. 19.35% by weight of a pigment dispersion (Jetsperse® AQ White (AQD-PWHT)) (62% solids) b. 19.37% by weight of a polymer dispersion (Takelac™ WPB-341) (30% solids) c. 17.43% by weight of a co-solvent (3-methoxy-3-methyl-1-butanol) d. 0.30% by weight of a surfactant (BYK® 345) e. 43.55% by weight of deionized water (DI H₂O)

(339) The 43.55% DI H₂O is additional deionized water to the water already used for The dispersions, such as Takelac™ WPB-341 dispersion. The total amount of deionized water throughout the printing composition is 57.12% (counting dispersion and other subcomponent water content).

(340) Example 4. A printing composition according to the present disclosure via ink jet application has the following formulation by weight: a. 11.20% by weight of a pigment dispersion (FujiFilm APD 1000 Cyan) (15% solids) b. 35.00% by weight of a polymer solution (Lawter™ Filtrez™ 526A modified rosin ester) (32% solids) c. 15.00% by weight of a co-solvent (3-methoxy-3-methyl-1-butanol) d. 1.00% by weight of a surfactant (BYK® 345) e. 37.80% by weight of deionized water (DI H₂O)

(341) The 37.80% DI H₂O is additional deionized water to the water already used for the dispersions. The Lawter™ Filtrez™ 526A modified rosin ester had a pH of 9.00. At 25° C., the printing composition has a viscosity of 4.57 cP, a surface tension of 31.0 dynes/cm, and pH of 8.92.

(342) Example 5. A printing composition according to the present disclosure by ink jet application has the following formulation by weight: a. 30.00% by weight of a pigment dispersion (FujiFilm APD 1000 Magenta) (15% solids) b. 29.00% by weight of a polymer solution (Lawter™ Filtrez™ 526A modified rosin ester) (32% solids) c. 15.00% by weight of a co-solvent (3-methoxy-3-methyl-1-butanol) d. 1.00% by weight of a surfactant (BYK® 345) e. 25.00% by weight of

deionized water (DI H₂O)

(343) The 25.00% DI H₂O is additional deionized water to the water already used for the dispersions. The Lawter™ Filtrez™ 526A modified rosin ester had a pH of 9.00. At 25° C., the printing composition has a viscosity of 4.64 cP, a surface tension of 31.2 dynes/cm, and pH of 8.90.

(344) Example 6. A printing composition according to the present disclosure by ink jet application has the following formulation by weight: a. 30.00% by weight of a pigment dispersion (FujiFilm APD 1000LF Yellow) (15% solids) b. 22.50% by weight of a polymer solution (Lawter™ Filtrez™ 526A modified rosin ester) (32% solids) c. 15.00% by weight of a co-solvent (3-methoxy-3-methyl-1-butanol) d. 1.00% by weight of a surfactant (BYK® 345) e. 31.50% by weight of deionized water (DI H₂O)

(345) The 31.50% DI H₂O is additional deionized water to the water already used for the dispersions. The Lawter™ Filtrez™ 526A modified rosin ester had a pH of 9.00. At 25° C., the printing composition has a viscosity of 4.56 cP, a surface tension of 31.8 dynes/cm, and pH of 8.94.

(346) Example 7. A printing composition according to the present disclosure by ink jet application has the following formulation by weight: a. 30.00% by weight of a pigment dispersion (FujiFilm APD 1500 Black) (15% solids) b. 25.50% by weight of a polymer solution (Lawter™ Filtrez™ 526A modified rosin ester) (32% solids) c. 15.00% by weight of a co-solvent (3-methoxy-3-methyl-1-butanol) d. 1.00% by weight of a surfactant (BYK 345) e. 28.50% by weight of deionized water (DI H₂O)

(347) The 28.50% DI H₂O is additional deionized water to the water already used for the dispersions. The Lawter™ Filtrez™ 526A modified rosin ester had a pH of 9.00. At 25° C., the printing composition has a viscosity of 4.50 cP, a surface tension of 31.4 dynes/cm, and pH of 8.87.

(348) Example 8. A printing composition useful in the present disclosure for flexographic application comprises the following formulation by weight: a. 30.00% by weight of a viscosity modifier dispersion (Methocel™ A15LV) (5% solids) b. 10.00% by weight of a polymer dispersion (Vinnapas® 410) (55% solids) c. 1.00% by weight of a surfactant (Surfynol® 465) d. 1.50% by weight of a defoamer agent dispersion (Surfynol® DF 110L) (20% solids) e. 0.50% by weight of a surface additive (BYK®-333) f. 57.00% by weight of deionized water (DI H₂O)

(349) Example 9. A printing composition useful in the present disclosure for flexographic application comprises the following formulation by weight: a. 30.00% by weight of a viscosity modifier dispersion (Methocel™ A15LV) (5% solids) b. 20.00% by weight of a polymer dispersion (Vinnapas® 410) (55% solids) c. 1.00% by weight of a surfactant (Surfynol® 465) d. 0.80% by weight of a defoamer agent dispersion BYK®-023) (19% solids) e. 5.00% by weight of a wax additive dispersion (Ultralube® E-912) (35% solids) f. 5.00% by weight of a polymer modifier dispersion (Rheolate® 212) (20% solids) g. 1.00% by weight of a polyether siloxane copolymer (TEGO® Glide 410) h. 37.20% by weight of deionized water (DI H₂O)

(350) The 37.20% DI H₂O is additional deionized water to the water already used for other. The total amount of deionized water throughout the printing composition is 82.60% (counting dispersion and other subcomponent water content).

INDUSTRIAL APPLICABILITY

(351) In summary, the web handling system **200** utilizes one or more precisely grooved nip rollers, multiple cross-grooved idler rollers, accurately aligned and spaced contact points throughout the entire system, and dynamic splice detection and subsequent image head retraction to minimize the possibility of wrinkle formation and damage therefrom. In addition, spreader rollers before important imaging nip points and dynamic gap and tension control also help minimize risk of system and product damage. The control system is operable to undertake closed-loop printhead gapping, splice, and/or wrinkle detection and printhead retraction to prevent printhead damage.

(352) Also in summary, a.) tension measurements of the previous zone are used to adjust driven rollers to achieve closed loop control; b.) control calculations allow for a wide range of change but at a slower rate to build tension in the elastic plastic film; and c.) multiple PID control algorithms

are used for each tension control (i.e., driven) roller comprising a first PID controller tuned to control roller positions relatively quickly to maintain synchronized movement of all rollers, and a second PID controller responsive to tension feedback for each zone that adjusts roller positions relatively slowly.

(353) Further, the system **20** including the control system **130** adjusts the registration from imager unit to imager unit without using any mechanical adjustment. The digital system **130** adjusts the firing of the printheads without the need to move the substrate or the print head array for registration purposes. By not moving the web around laterally, wrinkles are controlled/eliminated.

(354) The print system **20** also allows for dual side printing using multiple imager units on a single print drum per imager unit. Also, each imager unit can be virtually/digitally decoupled, so each portion of each imager unit **30**, **44**, **60**, **70**, and/or **82** can print independently from the other. Registration alignment can be made from imager unit to imager unit, side to side and back to front. This alignment can be processed through a camera and/or a high-speed sense mark system.

(355) Further, in some embodiments, the printing system **20** utilizes one or more printing methodologies such as flexography and ink jet to deposit a primer improving the ability to print an ink jet image on a flexible/shrinkable and/or an impermeable substrate, such as a heat-shrinkable substrate that is continuously variable at a high printing throughput speed with coating(s) and/or ink(s) that are water-based and comprise food law compliant component(s) for food packaging, such as substance(s) listed in annex(es) of Swiss Ordinance RS 817.023.21.

(356) In addition, in some embodiments, the printing system **20** for printing on a clear polymeric film web **24** includes a plurality of imager units **44**, **60**, **82**, and **228** having inkjet print units **1184**, **1186**, **1188**, and **228**. Such inkjet print units are operated to deposit material on the web **24** at a particular resolution and drop size selected in accordance with the type of material being deposited by such inkjet print unit and content being reproduced thereby. In addition, a distortion corrector **1200** determines adjustments to the dimensions and position on the web **24** of a page element prior being printed to compensate for distortion of the printed page element that results from heat shrinking of a bag manufactured from the web **24** around a product disposed therein.

(357) Also, a dryer management system **2217** is disclosed herein that operates on or more dryer unit(s) **32**, **46**, **64**, **80**, and/or **84** to dry material disposed on a web. It should be apparent to one who has ordinary skill in the art that the embodiments of the dryer management system **2217** disclosed herein may be adapted to dry any type of material deposited on any type of substrate using heat and/or a flow of heated air. Further, it should be apparent such embodiments may be adapted to dry material deposited on a substrate using any type of material deposition process.

(358) All references, including publications, patent applications, and patents, cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

(359) The use of the terms “a” and “an” and “the” and similar references in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the disclosure and does not pose a limitation on the scope of the disclosure unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the disclosure.

(360) Numerous modifications to the present disclosure will be apparent to those skilled in the art in view of the foregoing description. It should be understood that the illustrated embodiments are

exemplary only, and should not be taken as limiting the scope of the disclosure. This written description uses examples to disclose the invention, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A printing system, comprising: a transport apparatus adapted to transport a flexible web along a process direction; first and second individually controllable ink jet imager units offset from one another along the process direction wherein each of the first ink jet imager unit and the second ink jet imager unit includes a first portion operable to print on a first portion of the web and a second portion operable to print on the second portion of the web wherein each of the first portion and second portion of the first and second ink jet imager units is stationary along the process direction and the lateral direction and wherein the first ink jet imager unit prints on the web at a first print resolution and the second ink jet imager unit prints on the web at a second print resolution different than the first print resolution; the second ink jet imager unit comprising a plurality of inkjet printing units disposed around a circumference of a drum; each inkjet printing unit comprising a plurality of inkjet printheads such that ejection nozzles of the inkjet printheads substantially span the width of the flexible web; a first closed-loop control system that controls web handling and a distance of at least one of the first portion and the second portion of at least one of the first ink jet imager unit and the second ink jet imager unit from the web; and a second closed-loop control system that controls registration between the first ink jet imager unit and the second ink jet imager unit.
2. The printing system of claim 1, wherein the first closed-loop control system includes a printhead position sensor and a printhead actuator.
3. The printing system of claim 1, further including a plurality of cross-grooved idler rollers that support the web wherein each cross-grooved idler roller comprises two independently journaled portions.
4. The printing system of claim 3, further including an imaging drum, a nip roller adjacent the imaging drum and forming a nip therewith, and a spreader roller disposed between the plurality of cross-grooved idler rollers and the nip roller.
5. The printing system of claim 4, wherein each idler roller is disposed no more than a first distance from an adjacent idler roller, spreader roller, and nip roller and wherein the spreader roller is disposed no more than a second distance from the nip, wherein the first distance is in a range of 38 to 28 inches, 36 to 30 inches, or 35 to 33 inches and the second distance is in a range of 6 to 11 inches, 6.5 to 9 inches, or 7.0 to 8.5 inches.
6. A printing system for printing a heat-shrinkable web, comprising: a transport apparatus adapted to transport a heat shrinkable flexible web along a process direction, the heat-shrinkable flexible web having a total free shrink at 185° F. of at least 20% as measured by ASTM D2732; first and second individually controllable ink jet imager units offset from one another along the process direction wherein each of the first ink jet imager unit and the second ink jet imager unit includes a first portion operable to print on a first portion of the web and a second portion operable to print on the second portion of the web wherein each of the first portion and second portion of the first and second ink jet imager units is stationary along the process direction and the lateral direction and wherein the first ink jet imager unit includes printheads that print on the web at a first print resolution and the second ink jet imager unit includes printheads that print on the web at a second

print resolution different than the first print resolution; a first closed-loop control system that controls web handling and a distance of at least one of the printheads from the web; a second closed-loop control system responsive to web temperature that controls print drying without substantial shrinking of the web; and a third closed-loop control system that controls registration between the first ink jet imager unit and the second ink jet imager unit.

7. The printing system of claim 6, wherein the first closed-loop control system includes a printhead position sensor and a printhead actuator.

8. The printing system of claim 6, further including a plurality of cross-grooved idler rollers that support the web wherein each cross-grooved idler roller comprises two independently journaled portions.

9. The printing system of claim 8, further including an imaging drum, a nip roller adjacent the imaging drum and forming a nip therewith, and a spreader roller disposed between the plurality of cross-grooved idler rollers and the nip roller.

10. The printing system of claim 9, wherein each idler roller is disposed no more than a first distance from an adjacent idler roller, spreader roller, and nip roller and wherein the spreader roller is disposed no more than a second distance from the nip, wherein the first distance is in a range of 38 to 28 inches, 36 to 30 inches, or 35 to 33 inches and the second distance is in a range of 6 to 11 inches, 6.5 to 9 inches, or 7.0 to 8.5 inches.

11. A printing system for printing a heat-shrinkable web, comprising: a transport apparatus adapted to transport a flexible heat-shrinkable web along a process direction, the heat-shrinkable flexible web having a total free shrink at 185° F. of at least 20% as measured by ASTM D2732; first and second individually controllable ink jet imager units offset from one another along the process direction wherein each of the first ink jet imager unit and the second ink jet imager unit includes a first portion operable to print on a first portion of the web and a second portion operable to print on the second portion of the web wherein each of the first portion and second portion of the first and second ink jet imager units is stationary along the process direction and the lateral direction; at least one dryer operable to dry print on the web without substantial shrinking of the web; a first closed-loop control system responsive to an indication of web temperature that controls the at least one dryer; and a second closed-loop control system that controls registration between the first ink jet imager unit and the second ink jet imager unit.

12. The system of claim 11, further including a distortion corrector that adjusts dimensions of a page element to be printed on the web to compensate for distortion of the printed page element due to shrinking of the web.

13. The system of claim 12, wherein the distortion corrector includes a customer-site distortion analyzer that analyzes an image of the web having a product disposed therein to adjust distortion parameters.

14. The system of claim 11, wherein the at least one dryer includes a heater unit associated with and disposed downstream of one of the ink jet imager units and adapted to generate a flow of heated air to heat the web.

15. The system of claim 14, further comprising a camera disposed downstream of the heater unit, wherein the first closed-loop control system receives an image of the web from the camera, analyzes the received image to detect insufficient drying of material deposited on the web, and in response adjusts operation of the heater unit.

16. The system of claim 11, wherein the second closed-loop control system is adapted to register first content printed by the first portion of the first ink jet imager unit with content printed by the first portion of the second ink jet imager unit, register content printed by the second portion of the first ink jet imager unit with content printed by the second portion of the second ink jet imager unit, independently control the first portion and the second portion of the first ink jet imager unit, and independently control the first portion and the second portion of the second ink jet imager unit.

17. A printing system for printing a heat-shrinkable web, comprising: a transport apparatus adapted

to transport a flexible heat-shrinkable web along a process direction, the heat-shrinkable flexible web having a total free shrink at 185° F. of at least 20% as measured by ASTM D2732; first and second individually controllable ink jet imager units offset from one another along the process direction wherein each of the first ink jet imager unit and the second ink jet imager unit includes a first portion operable to print on a first portion of the web and a second portion operable to print on the second portion of the web wherein each of the first portion and second portion of the first and second ink jet imager units is stationary along the process direction and the lateral direction and wherein the first ink jet imager unit prints on the web at a first print resolution and the second ink jet imager unit prints on the web at a second print resolution different than the first print resolution; at least one dryer operable to dry print on the web without substantial shrinking of the web; a first closed-loop control system responsive to an indication of web temperature that controls the at least one dryer; and a second closed-loop control system that controls registration between the first ink jet imager unit and the second ink jet imager unit wherein the second closed-loop control system is adapted to register first content printed by the first portion of the first ink jet imager unit with content printed by the first portion of the second ink jet imager unit, register content printed by the second portion of the first ink jet imager unit with content printed by the second portion of the second ink jet imager unit, independently control the first portion and the second portion of the first ink jet imager unit, and independently control the first portion and the second portion of the ink jet second imager unit.

18. The system of claim 17, further including a distortion corrector that adjusts dimensions of a page element to be printed on the web to compensate for distortion of the printed page element due to shrinking of the web.

19. The system of claim 18, wherein the distortion corrector includes a customer-site distortion analyzer that analyzes an image of the web having a product disposed therein to adjust distortion parameters.

20. The system of claim 17, wherein the at least one dryer includes a heater unit associated with and disposed downstream of one of the ink jet imager units and adapted to generate a flow of heated air to heat the web.

21. The system of claim 20, further comprising a camera disposed downstream of the heater unit, wherein the first closed-loop control system receives an image of the web from the camera, analyzes the received image to detect insufficient drying of material deposited on the web, and in response adjusts operation of the heater unit.
